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Numerical aspects of the crack band approach

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ABSTRACT

The crack band approach has been used in many engineering computations as a simple technique eliminating or reducing the sensitivity of numerical results to the size of finite elements in simulations that involve strain localization due to softening. In the present paper, this traditional technique is carefully analyzed and potential sources of errors are identified. The role of various components of the modeling strategy is elucidated by simple examples dealing with a one-dimensional tensile test and a two-dimensional model of a notched beam subjected to bending. The influence of the specific choice of finite elements, integration scheme and formula for estimation of the crack band width is analyzed. Particular attention is devoted to the differences among band width estimation methods based on element area or volume, on element projection, and on the idea proposed by Oliver [32]. The results are summarized in the form of recommendations which can contribute to reduction of errors in practical simulations.

1. Introduction

The effect of propagating microscopic defects (such as cracks or voids) on the mechanical behavior of a material can be conveniently described within the framework of continuum damage mechanics. The growth and coalescence of such defects may lead to softening on the macroscopic level, i.e., to decreasing stress at increasing strain. Softening is one of the destabilizing factors that can induce localization of dissipative processes into narrow bands, the thickness of which is related to the size and spacing of material heterogeneities. Within the framework of the standard continuum with local action, the description of localized failure suffers from serious theoretical deficiencies, which result in pathological sensitivity of the numerical solution to the discretization parameters, such as the size of finite elements [1].

Objectivity of the description can be restored by various sophisticated enhancements of the underlying continuum theory, e.g. by incorporation of displacement discontinuities [2,3], or by nonlocal formulations of different kinds [4–13]. A simple remedy, frequently used in practical applications, can be based on an appropriate adjustment of the constitutive law depending on the width of the numerically resolved localized band, which is closely related to the finite element size [14,15]. Traditionally, this remedy is referred to as the crack band approach, fracture energy trick, or mesh-dependent softening modulus.

The adjustment of the constitutive model applied after the onset of localization consists in a rescaling of the post-localized part of the stress–strain law. The main objective is to preserve the overall energy dissipated by the failure process. The area under the stress–strain diagram, which represents dissipated energy per unit volume, has to be adjusted in inverse proportion to the width of the localized failure band. The product of the dissipation density (per unit volume) and the band width represents the fracture energy, i.e., the energy dissipated per unit area of the macroscopic crack surface which is approached in the limit as the band width tends to zero.

In the present contribution, the role of various components of a modeling strategy based on the crack band approach is discussed and several potential sources of error are revealed and analyzed. Attention is focused on a specific class of material models with softening, namely isotropic damage models with a strain-driven scalar damage variable. The following factors influencing the energy dissipated in the failure process are investigated:

- shape of the failure envelope in the stress space,
- technique used for rescaling of the stress–strain diagram,
- element type (triangles or quadrilaterals),
- order of the displacement approximation (linear or quadratic),
- alignment of the crack band with the finite element mesh,
- integration scheme (number of integration points, full versus selectively reduced integration),
- formula for the estimation of the crack band width.

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For simplicity, only one specific class of material models is considered, namely the isotropic damage models with one damage variable driven by the equivalent strain. Models of this kind are sufficiently representative for the present purpose. Of course, the inelastic response of cracking materials could be described by more refined models that take into account the cracking induced anisotropy, such as the rotating or fixed crack models, various anisotropic damage models, or microplane models. Even in the context of isotropic damage, more refined formulations with multiple damage variables or with stress-based loading–unloading conditions could be considered. However, the focus here is on proper treatment of the mesh-size dependence and the general trends observed for the simple isotropic damage models would remain the same for other constitutive frameworks.

All numerical simulations have been performed with OOFEM, an open-source finite element package developed by Patzák and coworkers [16,17], using extensions implemented by the present authors.

2. Isotropic damage model

A simple class of isotropic damage models is based on the stress–strain equation in the form

$$\boldsymbol{\sigma} = (1-\omega)\boldsymbol{D}_{\mathrm{e}} : \boldsymbol{\varepsilon} \tag{1}$$

where $\boldsymbol{\sigma}$ is the stress tensor, $\boldsymbol{\varepsilon}$ is the strain tensor, $\boldsymbol{D}_{\mathrm{e}}$ is the elastic stiffness tensor, and ω is a scalar damage variable, evolving from 0 for the intact material to 1 for the fully damaged material. In the strain-driven version of the model, the damage variable explicitly depends on the maximum previously reached level of a scalar measure of strain, called the equivalent strain, $\tilde{\varepsilon}$. The maximum level of equivalent strain, denoted as κ, plays the role of an internal variable and is formally described by the loading–unloading conditions

$$f(\boldsymbol{\varepsilon},\kappa) \leqslant 0, \quad \dot{\kappa} \geqslant 0, \quad \dot{\kappa} f(\boldsymbol{\varepsilon},\kappa) = 0 \tag{2}$$

with the damage loading function defined as

$$f(\boldsymbol{\varepsilon},\kappa) = \tilde{\varepsilon}(\boldsymbol{\varepsilon}) - \kappa \tag{3}$$

The specific choice of the expression for equivalent strain is related to the shape of the elastic domain in the strain space. For instance, setting $\tilde{\varepsilon}$ equal to the norm of the strain tensor, $\|\boldsymbol{\varepsilon}\|$, one would obtain a model with the same behavior in tension and in compression, which is not suitable for quasibrittle materials with damage due to cracking. A popular formula for equivalent strain, proposed by Mazars [18], is based on the norm of the positive part of the strain tensor:

$$\tilde{\varepsilon}(\boldsymbol{\varepsilon}) = \|\langle\boldsymbol{\varepsilon}\rangle\| = \sqrt{\sum_{I=1}^{3}\langle\varepsilon_I\rangle^2} \tag{4}$$

The Macauley brackets $\langle\cdots\rangle$ denote the positive part, and ε_I, $I = 1, 2, 3$, are the principal strains.

An alternative formula, referred to as the modified von Mises equivalent strain definition [6], reads

$$\tilde{\varepsilon}(\boldsymbol{\varepsilon}) = \frac{(k-1)I_1(\boldsymbol{\varepsilon})}{2k(1-2\nu)} + \frac{1}{2k}\sqrt{\frac{(k-1)^2}{(1-2\nu)^2}I_1^2(\boldsymbol{\varepsilon}) + \frac{12kJ_2(\boldsymbol{\varepsilon})}{(1+\nu)^2}} \tag{5}$$

where I_1 is the first strain invariant (trace of the strain tensor), J_2 is the second deviatoric strain invariant, ν is the Poisson ratio, and k is a model parameter corresponding to the ratio between the uniaxial compressive strength f_{c} and uniaxial tensile strength f_{t}.

In problems with limited compressive stresses, it is possible to use a formula which leads to the same strength envelope as the Rankine yield condition in plasticity and therefore will be referred to as the Rankine equivalent strain:

$$\tilde{\varepsilon}(\boldsymbol{\varepsilon}) = \frac{1}{E}\max_{I=1,2,3}\bar{\sigma}_I(\boldsymbol{\varepsilon}) \tag{6}$$

Here, E is Young's modulus and $\bar{\sigma}_I$, $I = 1, 2, 3$, are the principal effective stresses, i.e., the eigenvalues of the effective stress tensor

$$\bar{\boldsymbol{\sigma}} = \boldsymbol{D}_{\mathrm{e}} : \boldsymbol{\varepsilon} \tag{7}$$

As a slight modification, the smooth Rankine equivalent strain is defined as the scaled norm of the positive part of effective stress:

$$\tilde{\varepsilon}(\boldsymbol{\varepsilon}) = \frac{\|\langle\boldsymbol{D}_{\mathrm{e}} : \boldsymbol{\varepsilon}\rangle\|}{E} = \frac{1}{E}\sqrt{\sum_{I=1}^{3}\langle\bar{\sigma}_I(\boldsymbol{\varepsilon})\rangle^2} \tag{8}$$

The scaling factor $1/E$ in (6) and (8) ensures that, under uniaxial tension, the equivalent strain is equal to the axial strain component. More details can be found in [19].

For illustration, Fig. 1 shows the biaxial strength envelopes constructed for the above-mentioned definitions of equivalent strain,

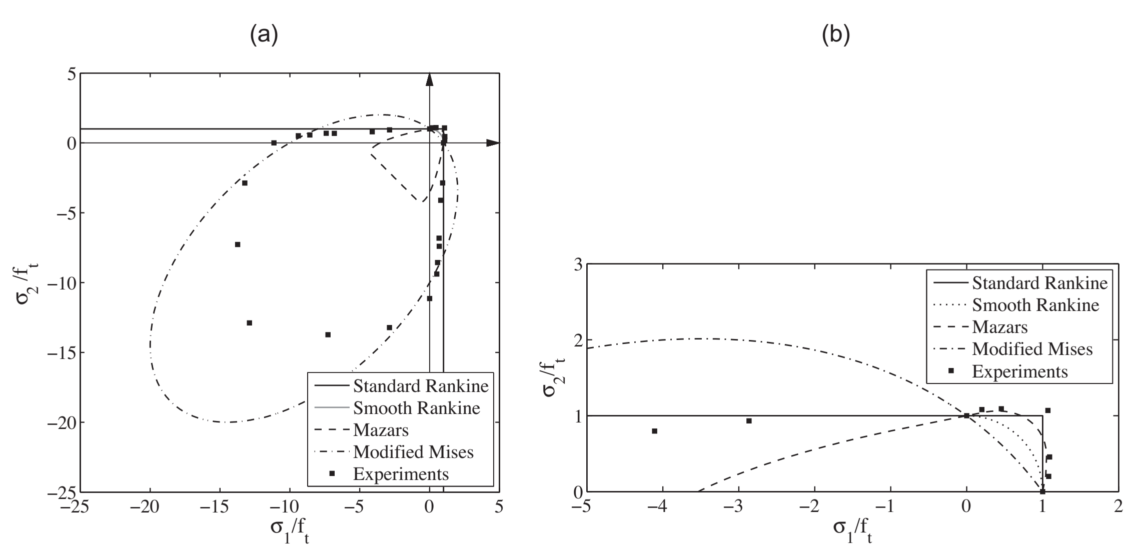

Fig. 1. Normalized biaxial strength envelopes corresponding to different definitions of equivalent strain: (a) global view and (b) detail.

using Poisson ratio $\nu = 0.2$ and, for the modified Mises definition, $k = 10$. These curves are plotted in the plane of principal stresses σ_1 and σ_2, with the out-of-plane principal stress σ_3 set to zero. Note that the out-of-plane principal strain is in general nonzero and can be expressed as

$$\varepsilon_3 = -\frac{\nu}{1-\nu}(\varepsilon_1 + \varepsilon_2) \tag{9}$$

which follows from the condition $\sigma_3 = 0$. For this reason, the Mazars equivalent strain is growing even under biaxial compression with $\sigma_1 = \sigma_2 < 0$ and $\sigma_3 = 0$, when $\varepsilon_1 = \varepsilon_2 < 0$ but $\varepsilon_3 > 0$. According to Mazars' definition, the strength envelope under plane stress is thus a closed curve (and the same holds for the modified Mises definition). On the other hand, for standard or smooth Rankine definitions the strength envelopes are unbounded in the region of uniaxial or biaxial compression. It is important to note that even though all the definitions give the same result under uniaxial tension, the influence of the transversal stress under biaxial conditions is quite different, as demonstrated in Fig. 1.

While the expression for equivalent strain controls the shape of the elastic domain in the strain or stress space, the shape of the stress–strain diagram is controlled by a function that describes the dependence of the damage variable on the internal variable κ (which is under monotonic loading equal to the equivalent strain $\tilde{\varepsilon}$). Softening curves of quasibrittle materials such as concrete are usually characterized by a relatively steep descent after the peak and by a long tail. This can be very well approximated by a special nonlinear law proposed in [20,21], or by a bilinear softening law with a low knee point [3,22,23].

For the present purpose, it is fully sufficient to consider a simple damage law in the form (Fig. 2b)

$$\omega = g(\kappa) \equiv \begin{cases} 0 & \text{for } \kappa \leqslant \varepsilon_0 \\ 1 - \frac{\varepsilon_0}{\kappa}\exp\left(-\frac{\kappa-\varepsilon_0}{\varepsilon_f-\varepsilon_0}\right) & \text{for } \kappa > \varepsilon_0 \end{cases} \tag{10}$$

which leads to a stress–strain diagram with linear elasticity up to the peak, followed by exponential softening (Fig. 2a):

$$\sigma(\varepsilon) = [1 - g(\varepsilon)]E\varepsilon = \begin{cases} E\varepsilon & \text{for } \varepsilon \leqslant \varepsilon_0 \\ E\varepsilon_0 \exp\left(-\frac{\varepsilon-\varepsilon_0}{\varepsilon_f-\varepsilon_0}\right) & \text{for } \varepsilon > \varepsilon_0 \end{cases} \tag{11}$$

Parameter ε_0 is the limit elastic strain (under uniaxial tension), and its product with Young's modulus E determines the tensile strength, $f_t = E\varepsilon_0$. Parameter ε_f controls the slope of the softening diagram, as indicated by the inclined dashed line in Fig. 2a (this line is tangent to the softening curve at peak and intersects the horizontal axis at ε_f).

The area under the stress–strain diagram,

$$g_F = \int_0^\infty \sigma(\varepsilon)\,\mathrm{d}\varepsilon = E\varepsilon_0(\varepsilon_f - \varepsilon_0/2) \tag{12}$$

represents the energy dissipated per unit volume (at complete failure). It is closely related to the fracture energy of the material, G_F, defined as the energy flux into a mode-I crack propagating in a stationary manner remote from the boundaries. For the standard cohesive crack model with a fixed traction-separation law (independent of the distance from the boundary, previous crack extension, normal stress components parallel to the crack, etc.), the fracture energy corresponds to the area under the traction-separation curve, which is equal to the dissipated energy per unit area of a completely broken ligament (under mode-I fracture conditions). For a continuum damage model with damage localized into a band of width h_b, we have $G_F = h_b g_F$.

3. Localization and crack band approach

It is well known that softening can induce localization of inelastic processes into zones of arbitrarily small thickness, which leads to a pathological sensitivity of the numerical results to discretization parameters such as the element size in finite element simulations. This can be overcome by regularization techniques based e.g. on integral-type nonlocal formulations [11] or various gradient-type enhancements [24,25]. However, practical engineering computations often exploit a simpler remedy, based on an appropriate adjustment of certain model parameters that control softening depending on the element size. The advantage is that the formulation remains local and the algorithmic structure of the finite element code requires only minor adjustments, limited to the part of the code responsible for evaluation of the stress (and stiffness) corresponding to a given strain increment. This family of techniques dates back to the pioneering work in the 1980s [14,15,26–33] and is in some form exploited in many finite element codes, including commercial ones. It has been presented under different names, such as the crack band approach, fracture energy trick, or mesh-adjusted softening modulus. For simplicity, we will use the term *crack band approach*, even though this type of approach is applicable not only to materials that experience cracking but also to other softening material models, e.g. to plasticity with degradation of yield stress.

As the most elementary localization test, consider a one-dimensional model of a prismatic homogeneous bar under uniaxial tension (with Poisson effect neglected). In the absence of body forces, static equilibrium conditions imply that the stress must

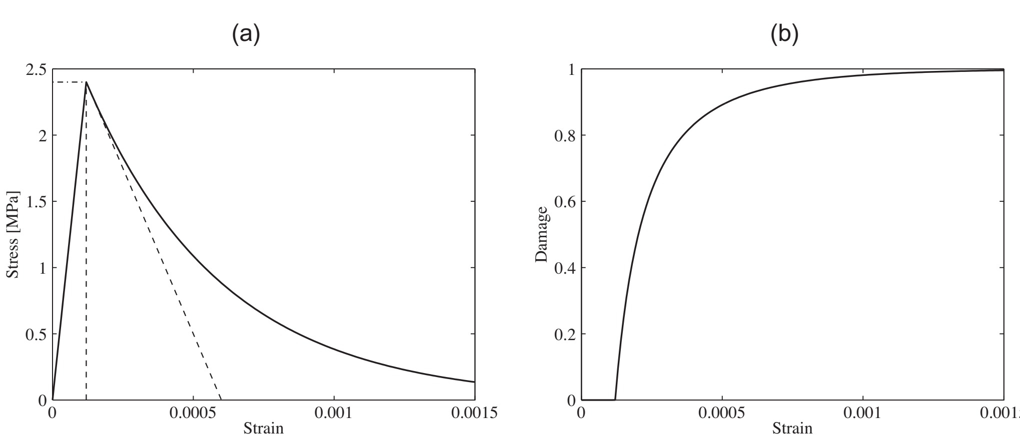

Fig. 2. (a) Stress–strain diagram with exponential softening and (b) the corresponding dependence of damage on strain.

be uniform along the whole bar. As long as the material response remains linear elastic (or hardening), the strain is uniquely determined by the stress, and the strain distribution remains uniform as well. When the stress state attains the peak of the stress–strain diagram, uniqueness of the solution is lost. In addition to the solution with uniform strain, there exist infinitely many other solutions for which the strain grows only in some parts of the bar while other parts are unloading. In particular, there exist solutions with damage growth localized into an interval of length h_b.

In general multi-dimensional numerical simulations, the size h_b of the numerically resolved localized damage band is related to the size, shape and orientation of finite elements, as will be discussed in detail in Sections 4 and 5. For the present purpose, let us assume that the values of h_b are known for different meshes. If the stress–strain law (11) was used with fixed parameters, the numerical results would exhibit pathological sensitivity to the mesh size. A simple remedy consists in an adjustment of certain model parameters depending on h_b. This is the general idea of the crack band approach.

Suppose that the bar under uniaxial tension has cross-sectional area A. If damage localizes into a band of size h_b, the volume of the softening material is Ah_b and the energy dissipated at complete failure is Ah_bg_F. To make sure that the dissipated energy per unit area is equal to the fracture energy G_F, considered as a material property, we must set $g_F = G_F/h_b$. This means that the stress–strain diagram is no longer considered as a unique curve characterizing the material but is adjusted according to the size of the localized damage band, h_b, which in turn depends on the finite element mesh. Since the elastic properties and the tensile strength should remain independent of the mesh, parameter $\varepsilon_0 = f_t/E$ must be kept constant, and only parameter ε_f can be adjusted. Eq. (12) with g_F replaced by G_F/h_b and $E\varepsilon_0$ replaced by f_t leads to

$$\varepsilon_f = \frac{\varepsilon_0}{2} + \frac{G_F}{h_b f_t} \tag{13}$$

Parameter ε_f increases with decreasing size of the localized band, h_b, and the corresponding stress–strain diagram becomes more ductile. On the other hand, for large values of h_b the adjusted stress–strain diagram exhibits dramatic softening. To prevent a snap-back at the material-point level (which would be highly questionable from the physical viewpoint and inadmissible from the computational viewpoint), parameter ε_f must not be smaller than ε_0. This condition leads to a restriction on the maximum size of the crack band:

$$h_b \leqslant h_{b,\max} = \frac{2G_F}{f_t\varepsilon_0} = \frac{2EG_F}{f_t^2} \tag{14}$$

The fraction EG_F/f_t^2 represents the characteristic material size [34], originally introduced by Hillerborg et al. [2], which is proportional to Irwin's estimate of the inelastic zone size in plastic (ductile) materials [35]. For concrete, it typically ranges between 150 mm and 400 mm.

As an example, consider concrete with E = 20 GPa, f_t = 2.4 MPa and G_F = 90 J/m^2. These parameters lead to a good fit of the experimental results in the case of three-point bending of a notched beam, to be simulated by a two-dimensional model in Section 5. The corresponding characteristic material size is $EG_F/f_t^2 = 312.5$ mm, and the maximum admissible size of the crack band according to (14) would be 625 mm. Fig. 3a shows the load–displacement diagram computed for a bar of length L = 450 mm and cross-sectional area A = 10,000 mm^2, using meshes consisting of one, three and nine equally sized finite elements with linear displacement interpolation, with damage localized in each case into a single element. The displacement on the horizontal axis is the total elongation of the bar. The area under all the load–displacement curves is the same and corresponds to the given fracture energy. However, the shape of the curve is sensitive to the number of elements, as long as the element size is comparable to the characteristic material size. For sufficiently small elements, the differences become negligible. This is documented by Fig. 3b, which shows the results for a bar of length L = 45 mm, again divided into one, three or nine equally sized finite elements. Of course, such a short concrete bar would in reality be difficult to test, but 45 mm can be considered as the gauge length on which the relative displacement is measured, rather than the total bar length.

For damage models with linear softening, the resulting global response of the one-dimensional finite element model of a bar under uniaxial tension would be totally independent of the element size. For the present model with exponential softening, such a perfect insensitivity to the mesh size could be achieved by a modified formulation, which splits the strain into the elastic part and the inelastic part and applies the size-dependent adjustment to the inelastic part only. This type of scaling of the stress–strain law is very natural and straightforward for those models that postulate the elastic and inelastic parts of the constitutive equations separately, such as elastoplastic models or smeared crack models. For damage models with nonlinear softening, it would lead to an implicit definition of the damage law and to the need for an iterative procedure; see [19] for further details and [36] for a related technique based on serial coupling with elastic elements. Since the typical size of the crack band in all subsequent simulations will be about 5 mm, such a refinement is not really needed here and the simple adjustment of parameter ε_f according to (13) is sufficient.

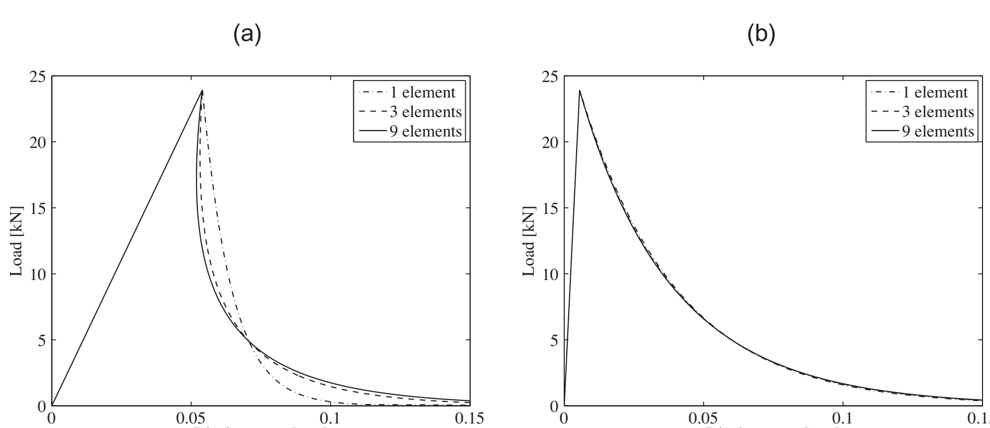

Fig. 3. Uniaxial load–displacement diagrams for a bar of length (a) 450 mm, (b) 45 mm, computed on different finite element meshes.

4. One-dimensional analysis

Let us perform a deeper analysis of the localized numerical solutions that are obtained in one-dimensional simulations of a bar under uniaxial tension. If the bar is discretized by *2-node finite elements with linear displacement interpolation*, the strain is uniform within each element and therefore the stress must also be uniform within each element. Discretized equilibrium conditions then imply that, in the absence of body forces, the stress must be uniform along the whole bar (which means that the equilibrium equation is satisfied at each point, i.e., in the strong sense). As long as the material response remains linear elastic (or hardening), the strain is uniquely determined by the stress, and the strain distribution remains uniform as well. When the peak of the stress–strain diagram is attained, uniqueness of the solution is lost and inelastic processes can localize into some parts of the bar.

The most localized solution, which minimizes energy dissipation, corresponds to softening in one element and unloading in all the other elements. Due to the piecewise constant nature of the strain approximation, the minimum size of the localized softening band is equal to the element size. Localization of softening into a shorter segment is excluded. Even if a numerical integration scheme with more than one Gauss point is used (which is of course not necessary in the one-dimensional case but could be meaningful in an analogous two-dimensional case), all the Gauss points of one element must behave in the same way and the response remains the same as for one-point integration.

The localization properties are substantially different for higher-order elements with non-constant strain approximation. For illustration, consider *3-node elements with quadratic displacement interpolation*. Fig. 4a shows two neighboring elements; one is assumed to experience softening and the other is elastically unloading. For one-point integration, spurious zero-energy modes would exist, and so the minimum number of Gauss integration points per element is two. For the two-point integration scheme, the Gauss points (marked by crosses) are located at distance $\xi h/2$ from the midpoint node (marked by a hollow circle), with h= element size and $\xi = 1/\sqrt{3}$, and their weights in the integration formula are the same, $w_1 = w_2 = 1/2$. It is easy to show that the discretized equilibrium equation associated with a midpoint node leads to the condition that the stresses be the same at both Gauss points (of the same element). If the Gauss points are numbered according to Fig. 4a, we can write this condition as

$$\sigma_1 = \sigma_2 \tag{15}$$

$$\sigma_3 = \sigma_4 \tag{16}$$

The discretized equilibrium equation associated with the node connecting the two elements (marked by a filled circle) reads

$$\sigma_1 + \sigma_2 = \sigma_3 + \sigma_4 \tag{17}$$

and combining this with conditions (15) and (16) we conclude that the stress in all Gauss points of these two elements (and by induction of all other elements) must be the same. One can expect a solution with softening localized into one element and with piecewise constant strain, analogous to the previous case with linear elements. Such a solution indeed exists, but it is not the most localized one. Softening can localize into one single Gauss point, which effectively represents a segment of the bar of length $h/2$. The corresponding strain distribution in the element is linear (Fig. 4b) and the stress distribution is even more complicated (Fig. 4c). It is clear that the equilibrium equation is no longer satisfied in the strong sense but it is satisfied in the weak sense provided that integrals are evaluated numerically using the two-point scheme. Therefore, a solution of this kind is indeed obtained in a numerical simulation, if the problem is perturbed and an imperfection is introduced, e.g. by slightly reducing the strength of one Gauss point.

Increasing the integration order does not help, and sub-element localization patterns still exist. For instance, for the three-point Gauss integration scheme, the integration points are located at the midpoint node and at distance $\zeta h/2$ from that node, with $\zeta = \sqrt{3/5}$, as shown in Fig. 5a. The corresponding integration weights are $w_2 = 8/18$ for the middle points and $w_1 = w_3 = 5/18$ for the other two points. The discretized equilibrium equations lead to the conditions

$$\sigma_1 = \sigma_3 \tag{18}$$

$$\begin{aligned}(1-2\zeta)w_1\sigma_1 + w_2\sigma_2 + (1+2\zeta)w_3\sigma_3 \\ = (1+2\zeta)w_1\sigma_4 + w_2\sigma_5 + (1-2\zeta)w_3\sigma_6\end{aligned} \tag{19}$$

$$\sigma_4 = \sigma_6 \tag{20}$$

and the linear approximation of strains in each element leads to the constraints

$$\varepsilon_1 - 2\varepsilon_2 + \varepsilon_3 = 0 \tag{21}$$

$$\varepsilon_4 - 2\varepsilon_5 + \varepsilon_6 = 0 \tag{22}$$

Since the right element with integration points 4, 5 and 6 is unloading, we have $\sigma_4 = E\varepsilon_4$, $\sigma_5 = E\varepsilon_5$ and $\sigma_6 = E\varepsilon_6$. Combining these equations with (20) and (22), we conclude that the strain and stress must be uniform within the unloading element, which is of course expected. In the left element, all Gauss points could be softening and then the solution would also be uniform within this element, but there exists another solution with softening localized into one or two Gauss points out of three (Fig. 5b). If Gauss point 1 is softening, we have $\sigma_1 = f_s(\varepsilon_1)$ where f_s is the function describing the softening branch of the stress–strain curve. The stress at Gauss point 3

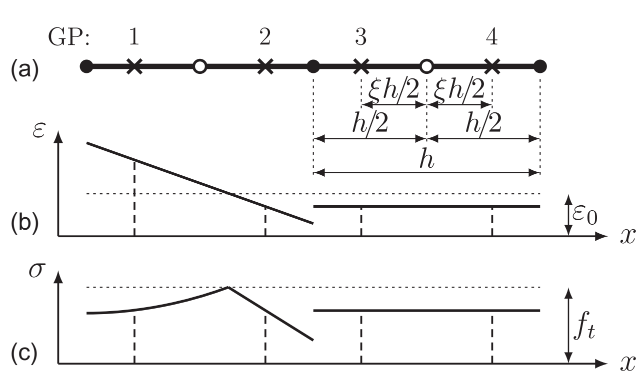

Fig. 4. (a) One-dimensional finite element mesh with quadratic elements and two-point Gauss integration, (b) strain distribution, and (c) stress distribution.

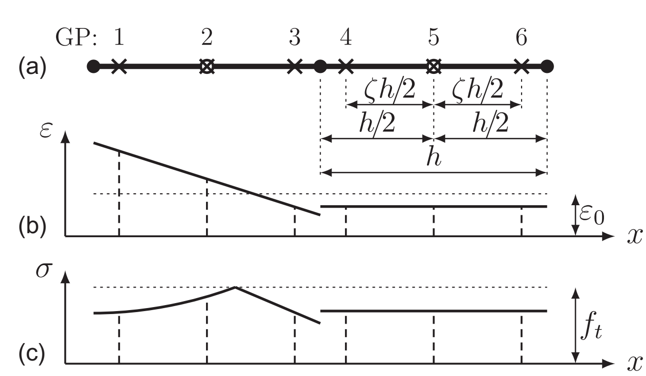

Fig. 5. (a) One-dimensional finite element mesh with quadratic elements and three-point Gauss integration, (b) strain distribution, and (c) stress distribution.

must be the same as at point 1, due to (18), but point 3 can be elastically unloading, with $\varepsilon_3 = \sigma_3/E = f_s(\varepsilon_1)/E$. The strain at point 2 is then computed from (21) and is given by

$$\varepsilon_2 = \frac{1}{2}(\varepsilon_1 + \varepsilon_3) = \frac{1}{2}\left(\varepsilon_1 + \frac{f_s(\varepsilon_1)}{E}\right) \tag{23}$$

If $\varepsilon_2 > \varepsilon_0$ (where ε_0 is the model parameter corresponding to the strain at peak), then point 2 is also softening and $\sigma_2 = f_s(\varepsilon_2)$, otherwise point 2 is unloading and $\sigma_2 = \varepsilon_2/E$. In either case, we obtain a solution localized into a segment shorter than one finite element. The stress in the neighboring unloading element (and in all other unloading elements), $\sigma_4 = \sigma_5 = \sigma_6$, is then evaluated from (19). In the case when $\varepsilon_2 > \varepsilon_0$ we get

$$\begin{aligned}\sigma_4 &= (1-2\zeta)w_1\sigma_1 + w_2\sigma_2 + (1+2\zeta)w_3\sigma_3 = 2w_1\sigma_1 + w_2\sigma_2 \\ &= 2w_1 f_s(\varepsilon_1) + w_2 f_s\left(\frac{\varepsilon_1}{2} + \frac{f_s(\varepsilon_1)}{2E}\right)\end{aligned} \tag{24}$$

From the stress–strain diagram in Fig. 6 with graphical interpretation of the above equations it is clear that if the softening curve is above the dashed straight line plotted from the peak with slope $-E$, then Gauss point 2 is softening simultaneously with Gauss point 1. If the softening curve is entirely below the dashed line (which could be the case e.g. for linear softening with negative tangent modulus that exceeds E in magnitude), point 2 is fully unloading and softening is localized into point 1 only. Finally, if the initial part of the softening curve is below the straight line but its tail is above that line, point 2 is first partially unloaded, later is reloaded to the peak again and then follows the softening curve. For all softening curves with complete failure at strains exceeding $2\varepsilon_0$ (including the exponential softening curve with complete failure at infinite strain), energy is

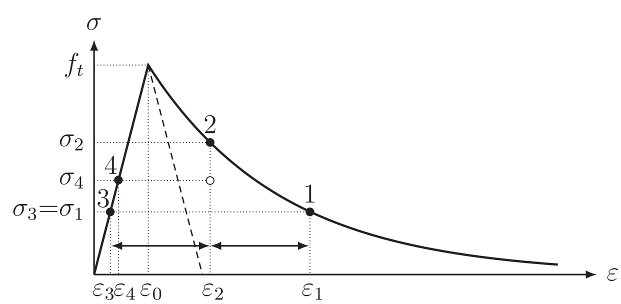

Fig. 6. Stress–strain diagram with the states at integration points 1–4 marked by filled circles and the average state in the element with softening (weighted average of points 1, 2 and 3) by an empty circle.

dissipated in two Gauss points with relative weight $(w_1 + w_2)/(w_1 + w_2 + w_3) = 5/18 + 8/18 = 13/18$, and the effective size of the localization zone, h_b, is thus 13/18 times the element size, h.

In summary, the foregoing analysis has revealed that the minimum size of the numerically resolved localized softening zone is not always equal to the size of one finite element. For elements with quadratic displacement interpolation, softening can localize into one Gauss point out of two, or into two (in some cases just one) Gauss points out of three. The effective size of the localized zone is then $h_b = h/2$ for two-point integration, and typically $h_b = 13h/18$ for three-point integration. This can be confirmed by numerical simulations.

Fig. 7 shows the load–displacement curves for a bar of total length $L = 45$ mm, divided into nine equally sized finite elements. Material properties were set again to $E = 20$ GPa, $f_t = 2.4$ MPa and $G_F = 90$ J/m^2. The reference solid curve in Fig. 7a represents the solution obtained with nine linear elements, using the crack-band approach with the band size set equal to the element size. Virtually the same results would be obtained for any number of linear elements, even on nonuniform meshes, provided that damage localizes into one finite element, sufficiently small with respect to the characteristic material length. For nine quadratic elements with two-point Gauss integration and with the crack band size set equal to the element size, the resulting behavior is more brittle; see the dashed curve in Fig. 7a. This is caused by the discrepancy between the assumed band size (equal to the element size) and the actual band size (equal to 1/2 of the element size). If the assumed band size is set to $h_b = h/2$, the results improve and the (almost) exact solution is obtained; see Fig. 7b. The dash-dotted curve in Fig. 7a corresponds to a simulation with nine quadratic elements using three-point Gauss integration and assumed band size set to the element size. Again, the response is too brittle, but not as much as for two-point integration. In this case, if the assumed band size is set to $h_b = 13h/18$, the resulting dash-dotted curve is close to but not identical with the exact one, as shown in Fig. 7b. The total dissipation (area under the curve) is reproduced correctly but the shape of the softening curve is slightly different. This is consistent with the results of our theoretical analysis.

5. Crack band in two dimensions

5.1. Reference case

Let us now proceed to an evaluation of the crack band approach in the two-dimensional setting. To be able to separate various ef-

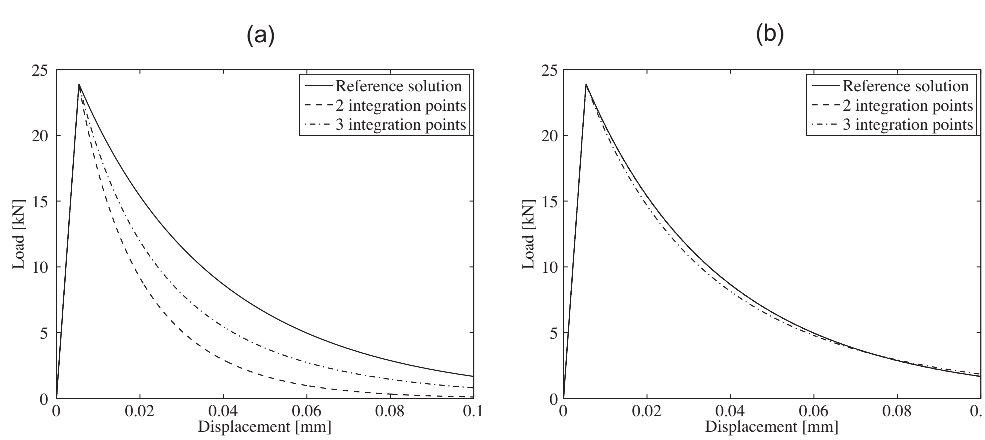

Fig. 7. Load–displacement curves for one-dimensional simulations of a bar under uniaxial tension using a mesh consisting of nine quadratic elements with two or three integration points: (a) assumed crack band size set to the element size, (b) assumed crack band size set to 1/2 of the element size for 2-point integration and to 13/18 times the element size for 3-point integration.

fects and clearly identify individual sources of errors, we consider a basic test with a very simple geometry, namely a notched beam subjected to three-point bending (Fig. 8a). The out-of-plane thickness is 100 mm. Concrete beams of this specific size and shape were investigated experimentally by Kormeling and Reinhardt [37] and simulated for instance in [31,38–41].

As the reference case, we consider the isotropic damage model with Rankine-type equivalent strain given by (6), a mesh consisting of bilinear quadrilateral elements with horizontal and vertical mesh lines (Fig. 8b) and 2×2 Gauss integration scheme. In the reference simulation, the width of the crack band is prescribed as $h_b = 5$ mm, which is the actual width of the vertical band of elements around the axis of symmetry. With material properties set to $E = 20$ GPa, $\nu = 0.2$, $f_t = 2.4$ MPa and $G_F = 90$ J/m^2, a good agreement with the experimental results reported in [37] is obtained; see Fig. 9. Note that the elastic constants and tensile strength have been taken by the same values as in [31] but the fracture energy has been reduced from 113 J/m^2 to the present value of 90 J/m^2. With the larger value, used e.g. in [31,39,41], the post-peak part of the load–displacement curve would be above the experimental range. Nevertheless, this is just a detail, since the present paper does not aim at optimized fitting of one particular set of experiments. The principal objective is to demonstrate the role of individual components of the simulation model and identify potential sources of errors, using a realistic set of model parameters.

In the subsequent simulations, the influence of individual model components will be assessed. We will consider various forms of the equivalent strain definition, various types of finite elements and integration schemes, and various rules for estimation of the crack band size. To limit the scope of the paper, all meshes will be constructed such that the overall direction of the crack band is captured correctly (Fig. 10). Mesh-induced directional bias will not be discussed here, and the analysis will be restricted to mode-I failure.

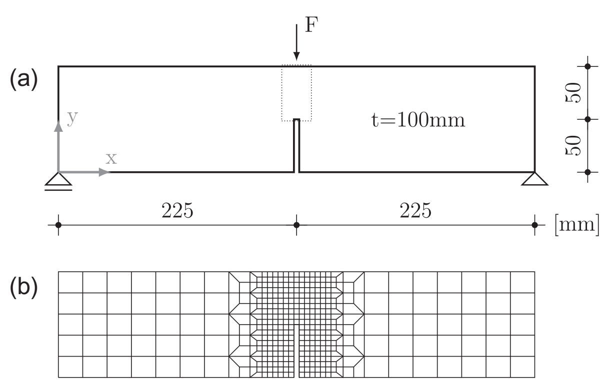

Fig. 8. Notched beam under three-point bending: (a) geometry and (b) reference mesh.

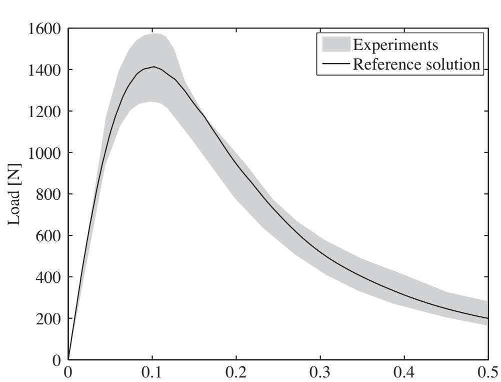

Fig. 9. Load–displacement curve obtained by the reference simulation and the range of experimental results from [37].

5.2. Influence of equivalent strain definition

Fig. 11 shows the influence of the choice of equivalent strain definition on the load–displacement diagram. All simulations have been run on the reference mesh (Fig. 8b) with the reference material parameters and with the assumed size of the crack band prescribed as $h_b = 5$ mm. Under uniaxial tension, the response would be independent of the definition of equivalent strain, because all the expressions are normalized such that, under uniaxial tension, they reduce to the axial strain. However, it turns out that the stress in the crack band which develops under bending is not really uniaxial. On the axis of symmetry of the beam, the dominating normal stress σ_x in the horizontal direction (parallel to the beam axis) is accompanied by a relatively high positive normal stress σ_y in the vertical direction, and the resulting stress state corresponds to biaxial tension. This is documented in Fig. 12, which shows the evolution of Cartesian stress components σ_x, σ_y and τ_{xy} (with respect to the global coordinate system from Fig. 8a) and of the in-plane principal stresses σ_1 and σ_2 at the Gauss point marked by a small filled circle in Fig. 10a. Since this point is not exactly on the axis of symmetry, the shear stress is nonzero, but it remains quite low compared to the normal stresses.

For the model with the standard Rankine definition of equivalent strain, Eq. (6), the peak value of the major principal stress σ_1 is exactly f_t. However, for the other definitions, the presence of a positive minor principal stress σ_2 reduces or increases the maximum tensile stress that can be transmitted by the material. This effect is related to the shape of the biaxial failure envelope. From Fig. 1 it can be deduced that the strength is reduced for the smooth Rankine definition (8) or modified Mises definition (5), is unaffected for the standard Rankine definition (6), and is increased for the Mazars definition (4), provided that the ratio σ_2/σ_1 does not exceed $(2 + \sqrt{3 - \nu^2})\nu/(1 + \nu^2)$, which is for $\nu = 0.2$ equal to 0.715. Indeed, this is in a good agreement with the effects observed on the structural level in Fig. 11a. The peak load obtained with the model based on Mazars' definition is by 5.3% higher than for the reference model with the standard Rankine definition, and the peak loads obtained with the models based on the smooth Rankine and modified Mises definitions are, respectively, by 3.8% and 18.4% lower.

The peak load under bending is affected not only by the material strength but also by the ductility, which can be characterized by the density of energy dissipated up to complete failure. Under uniaxial tension, the density of dissipated energy corresponds to the area under the stress–strain diagram. Under biaxial tension, the minor principal stress contributes to the dissipation as well, by the work done on the increments of the corresponding principal strain (note that, for isotropic damage models, the principal axes of stress and strain remain aligned). The density of dissipated energy corresponds to the sum of the areas under the diagrams of σ_1 versus ε_1 and σ_2 versus ε_2, which are shown in Fig. 13. Note the huge differences in scales, especially on the horizontal axes. The contribution of the major principal stress σ_1 to the dissipation is about 200 times larger than the contribution of the minor principal stress σ_2.

To get more insight, it is instructive to compute the density of the dissipative work (per unit volume) for proportional loading with different values of the biaxiality ratio σ_2/σ_1, which can be done analytically. The results are graphically presented in Fig. 14. They refer to mode-I fracture energy $G_F = 90$ J/m^2 and crack band

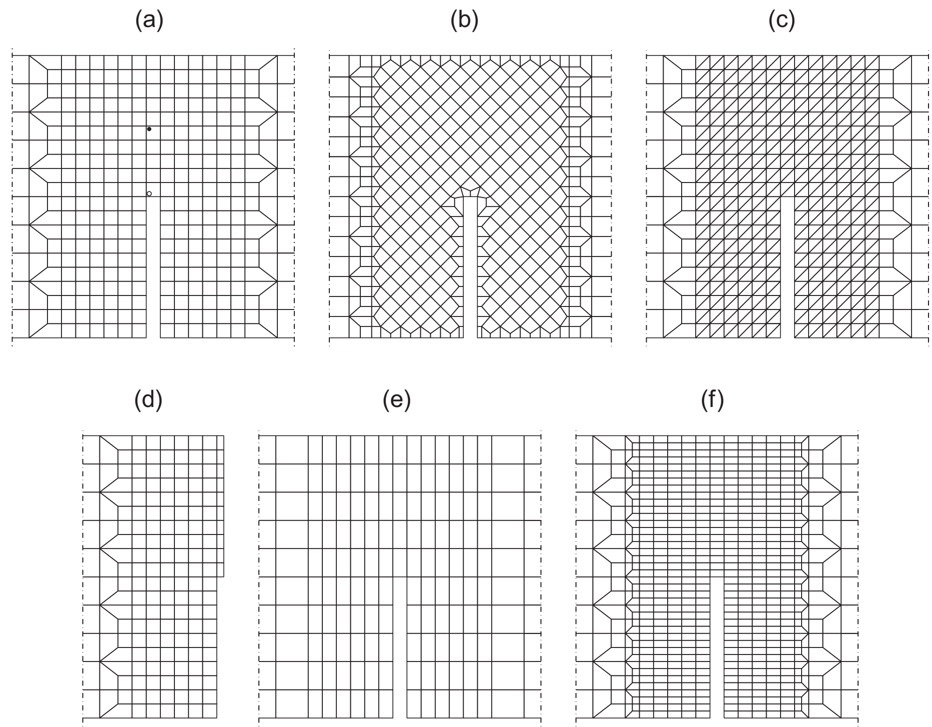

Fig. 10. Overview of finite element meshes to be used in the simulations: (a) reference mesh consisting of square elements with mesh lines aligned with the crack band direction, (b) square elements with mesh lines inclined with respect to the crack band direction, (c) triangular elements, (d) mesh adopted in simulations exploiting symmetry, and (e)–(f) elongated rectangular elements.

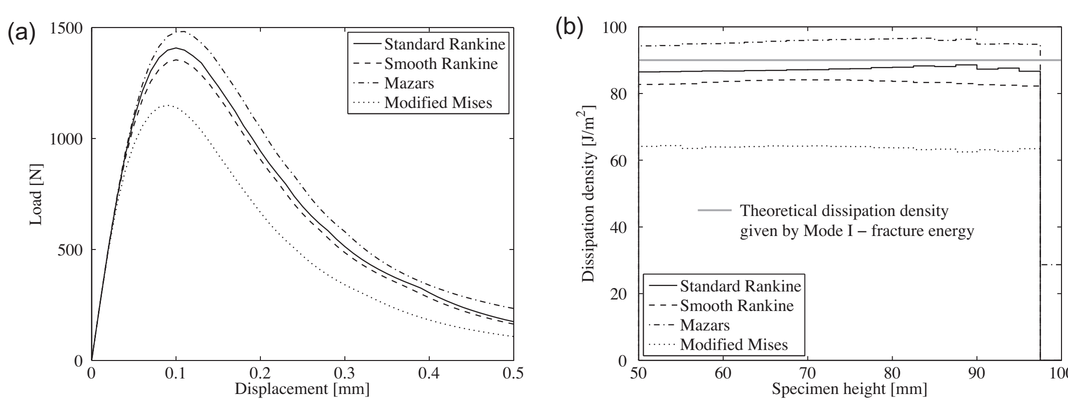

Fig. 11. (a) Load–displacement curves and (b) distribution of dissipated energy for models with different definitions of equivalent strain.

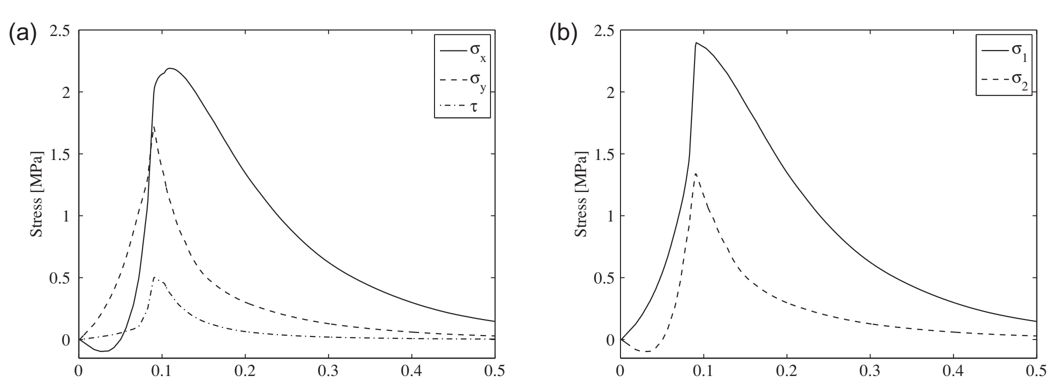

Fig. 12. Evolution of (a) Cartesian stress components and (b) principal stresses at Gauss point 3 of element 6.

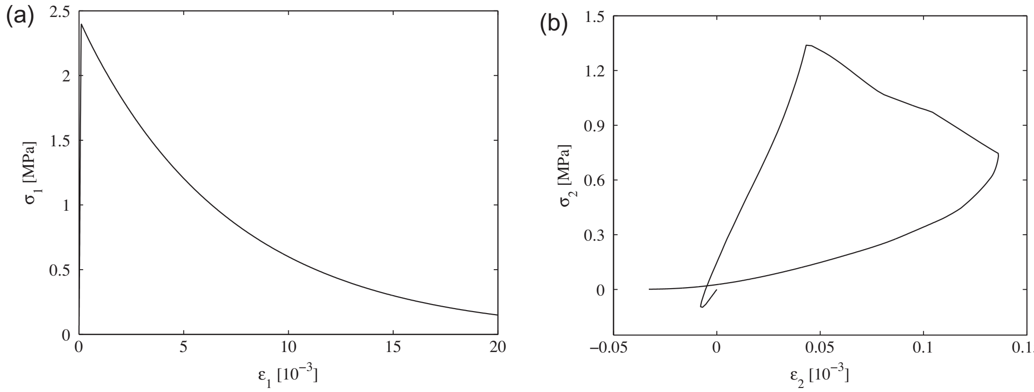

Fig. 13. Stress–strain diagrams under biaxial stress (with variable ratio of principal stresses σ_2/σ_1, plotted in terms of (a) major principal stress versus major principal strain and (b) minor principal stress versus minor principal strain.

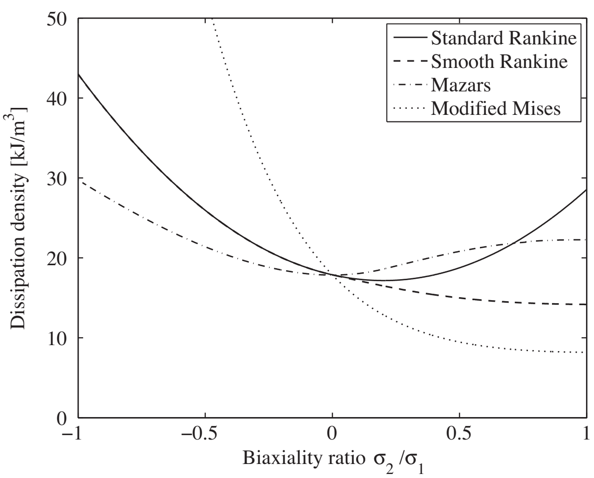

Fig. 14. Density of dissipative work under proportional loading, as a function of the biaxiality ratio σ_2/σ_1, for Poisson's ratio $\nu = 0.2$.

size $h_b = 5$ mm, for which the density of dissipative work under uniaxial tension is $g_F = G_F/h_b = 18$ kJ/m^3. With the proper adjustment of parameter $\varepsilon_f = 7.56 \times 10^{-3}$ according to (13), this value of g_F is exactly reproduced by all the models under uniaxial conditions, i.e., for $\sigma_2/\sigma_1 = 0$. As seen in Fig. 14, the dissipative work under biaxial tension with $0 < \sigma_2/\sigma_1 < 0.7$ is for the Mazars equivalent strain larger than for the standard Rankine, while for the smooth Rankine definition it is smaller and for the modified Mises definition it is much smaller. This explains the largely different areas under the load–displacement curves in Fig. 11a. To illustrate this effect in more detail, Fig. 11b shows the distribution of dissipated energy per unit area along the ligament (as a function of coordinate y, varying from 50 mm at the notch tip to 100 mm at the top surface of the beam).

To get an idea about the actual values of the principal stress ratio, the evolution of principal stresses at two selected Gauss points (marked in Fig. 10a by a filled circle and by a hollow circle) is presented in Fig. 15 in the form of trajectories in the principal stress plane. For reference, the figure also shows the biaxial strength envelopes for different definitions of equivalent strain. For the point marked by the filled circle, which is located approximately at mid-height of the ligament, the minor principal stress σ_2 is initially negative but at the onset of damage it is positive and approximately equal to 0.55 times the major principal stress σ_1 (Fig. 15a). During softening, the principal stress ratio σ_2/σ_1 decreases and eventually approaches 0.2. For the point marked by the hollow circle, which is located just above the notch, both principal stresses are always positive and their ratio decreases from approximately 0.32 at the onset of damage to 0.2 at late stages of softening (Fig. 15b). In the range of σ_2/σ_1 between 0.2 and 0.55, Fig. 14 indicates that the highest density of dissipative work is obtained for the Mazars model, then for the standard Rankine model, smooth Rankine model, and modified Mises model. This explains the differences observed in Fig. 11b.

Despite the large differences between the stress evolution at the two examined points, the principal stress ratio σ_2/σ_1 tends to 0.2 at late stages of softening, at both points and for all the models considered here. This is not just a coincidence. Inside the vertical crack band, the horizontal normal strain tends to infinity but the vertical normal strain tends to zero, because of the confining effect of the unloading material around the crack band. The ratio of the principal effective stresses computed from the condition $\varepsilon_2 = (\bar{\sigma}_2 - \nu\bar{\sigma}_1)/E = 0$ is $\bar{\sigma}_2/\bar{\sigma}_1 = \nu = 0.2$, and the ratio of the principal nominal stresses σ_2/σ_1 is the same. This is characteristic of isotropic damage models. For anisotropic models, σ_2 would typi-

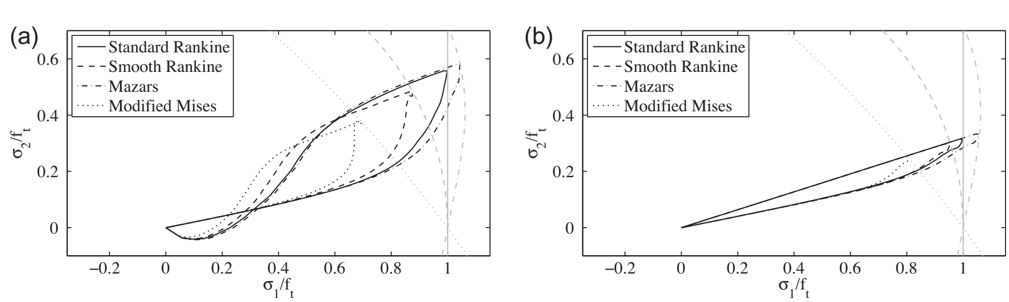

Fig. 15. Biaxial strength envelopes and the actual evolution of principal stresses at the Gauss points marked in Fig. 10a by (a) a filled circle and (b) a hollow circle.

cally tend to zero much faster than σ_1, and the stress state would be closer to uniaxial tension.

In the example of a notched beam considered here, the crack band experiences biaxial tension. If the normal stress parallel with the crack band is compressive, which happens e.g. in the Brazilian splitting test, the influence of the specific choice of the equivalent strain definition can be expected to be even more dramatic.

5.3. Criteria for estimation of crack band width

5.3.1. Criterion based on element area or volume

As already mentioned, in problems with strong softening the inelastic strain in general localizes into a band of elements running across the mesh and forming the so-called crack band. Sometimes it is recommended to estimate the crack band size as the square root of the element area (for two-dimensional elements) or the cubic root of the element volume (for three-dimensional elements). This rule, implemented in many commercial finite element packages, is easy to apply but it can induce a large error for elongated elements, and even for square or cube elements if the crack band is not aligned with the mesh. This is demonstrated in Fig. 16a, which shows the load–displacement curves obtained in simulations exploiting the square-root criterion. The dashed, dotted and dash-dotted curves refer to meshes with rectangular elongated elements, from Fig. 10e,f, and to the mesh from Fig. 10b, with square elements rotated such that the crack band is running along the diagonals. The solid curve is obtained on the mesh from Fig. 10a and agrees with the reference curve from Fig. 9 because, for this specific mesh, the square root of element area happens to be equal to the exact crack band width, h_b = 5 mm. It is obvious that the discrepancies are substantial and could be even more pronounced for meshes with more elongated and distorted elements.

For the triangular mesh from Fig. 10c, the load–displacement curves are presented in Fig. 16b. The precise form of the square-root criterion applicable to triangular elements is not obvious. For comparison, the simulation has been run with three different versions of that criterion. If the band width is set simply to $\sqrt{A}$, the response is too ductile, as indicated by the dash-dotted curve. However, for triangles it is hard to find a reasonable interpretation of $\sqrt{A}$ as the "element size". In analogy to the square as the most regular shape of a quadrilateral element, it seems logical to consider the equilateral triangle. The height of such a triangle of edge length h is $v = h\sqrt{3}/2$ and its area is $A = hv/2 = h^2\sqrt{3}/4$. If the crack band propagates in one layer of equilateral triangles, the band width is equal to the height of the triangle, $h_b = v$. In terms of the element area, the band width can then be expressed as $h_b = 3^{1/4}\sqrt{A} \approx 1.316\sqrt{A}$. The corresponding dotted curve in Fig. 16b is closer to the reference solution, but still above it. For the present mesh with triangular elements obtained by diagonal splitting of a square, the width of the band would best be evaluated as $h_b = \sqrt{2A} \approx 1.414\sqrt{A}$, and the corresponding dashed curve is indeed very close to the reference solution. However, it is clear that such a rule is ad hoc constructed for the specific mesh used in the present simulation and would not lead to an accurate estimate of the band width in a general situation. The square-root criterion is therefore excluded from further considerations.

5.3.2. Projection methods

The crack band is usually the smallest possible pattern that still allows separation of nodes on its opposite sides; see the shaded patterns in Fig. 17a–c. The correct value of the effective band size h_b is affected not only by the mesh size but also by the inclination of the crack band with respect to the mesh lines. This is illustrated by Fig. 17b, which explains why the correct value of h_b for a band

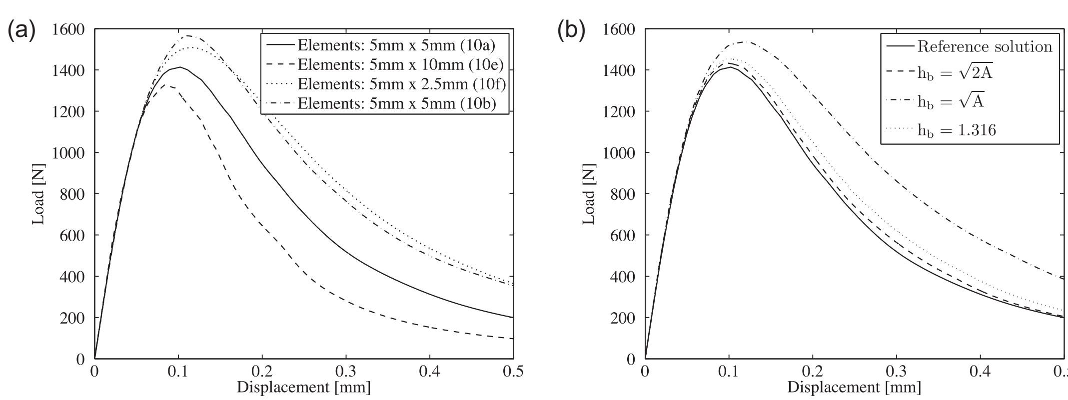

Fig. 16. Load–displacement curves for simulations with crack band width estimated as square root of element area: (a) bilinear quadrilateral elements (meshes from Fig. 10a, b, e, and f) and (b) linear triangular elements (mesh from Fig. 10c).

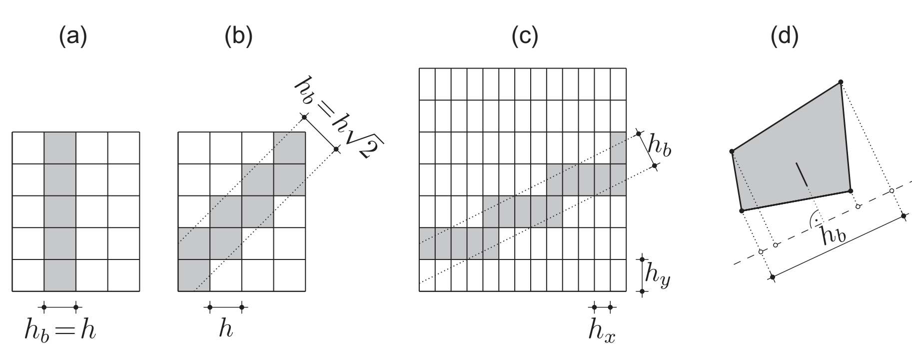

Fig. 17. (a) Crack band parallel to element sides, (b) crack band running along element diagonals, (c) inclined crack band in a rectangular mesh, and (d) determination of effective band size by projection.

propagating along the diagonals of a regular square mesh is $\sqrt{2}$ times larger compared to the case when the band is aligned with element sides (Fig. 17a). Suppose that the crack band propagates in a rectangular mesh in an inclined direction as shown in Fig. 17c. The area of the straight band between the dotted lines is the same as the area of the shaded elements that constitute the zig-zag crack band [27]. Therefore, the distance between the parallel dotted lines represents the effective band size, h_b. It is obvious that this distance corresponds to the projection of one element onto the direction perpendicular to the dotted lines.

This motivates a general rule that can be applied to elements of an arbitrary shape [42,43,34]: the effective crack band size is estimated by projecting the element onto the direction perpendicular to the assumed crack band direction, as shown in Fig. 17d. For efficiency, it is desirable to construct an accurate estimate of the overall crack band direction based on local information available within the given finite element. The simplest estimate can be based on the principal strain directions. The crack band is expected to be perpendicular to the direction associated with the major principal strain, and the effective crack band size is thus constructed by projecting the element onto that direction. In the examples in Section 6, we will examine the basic version of the projection method, with h_b determined at each Gauss point separately, and also the modified projection method, with one single value of h_b determined at the element center (i.e., based on the average strain in the element) and used at all Gauss points.

According to [43], the standard projection method leads to excessive dissipation in crack bands that are inclined with respect to the mesh lines. It was proposed to multiply the band width obtained by projection by a corrective orientation factor γ, which is equal to 1 for cracks perpendicular to the mesh lines, equal to 1.5 for cracks inclined by 45°, and interpolated in a linear fashion for general angles between the cracks and the mesh lines.

5.3.3. Methods proposed by Oliver

Yet another technique was developed by Oliver [32], based on an auxiliary function ϕ, which is set to 0 on one side of the crack band, set to 1 on the other side of the crack band, and interpolated across the crack band using the standard shape functions (for higher-order elements, only the shape functions associated with the corner nodes would be considered). The effective width of the crack band can be estimated as the reciprocal value of the directional derivative of function ϕ, evaluated in the direction normal to the band. Mathematically speaking,

$$h_b = \left(\frac{\partial\phi}{\partial\boldsymbol{x}}\cdot\boldsymbol{n}\right)^{-1} = \left(\sum_I \phi_I \frac{\partial N_I}{\partial\boldsymbol{x}}\cdot\boldsymbol{n}\right)^{-1} \tag{25}$$

where $\partial\phi/\partial\boldsymbol{x}$ is the gradient of function ϕ (evaluated at each Gauss point separately), $\boldsymbol{n}$ is the unit vector in the direction normal to the band, ϕ_I is the nodal value (equal to 0 or 1) and N_I is the shape function associated with corner node number I. The sum on the right-hand side of (25) is taken over all corner nodes of the element. The nodal values ϕ_I are assigned when the first Gauss point of the finite element under consideration attains the damage threshold. To determine the values of ϕ_I, the element is cut by a straight line passing through the element center and aligned with the expected crack band direction (perpendicular to the direction $\boldsymbol{n}_c$ associated with the major principal strain, evaluated at the element center). The auxiliary function ϕ is set to 0 at corner nodes on one side of this cutting line and to 1 on the other side; see Fig. 18a.

Strictly speaking, in [32] it was suggested to determine the expected crack band direction by localization analysis, based on the vanishing determinant of the so-called acoustic tensor, $\boldsymbol{n}\cdot\boldsymbol{D}_{\text{ed}}\cdot\boldsymbol{n}$, where $\boldsymbol{D}_{\text{ed}}$ is the tangent stiffness tensor of the damage model. As shown in [44], under biaxial tension with principal stress ratio $\sigma_2/\sigma_1 \geqslant \nu$ the discontinuity surface predicted by localization analysis of the isotropic damage model with (standard) Rankine definition of equivalent strain is indeed perpendicular to the principal direction associated with the major principal strain. This direction is characterized by a unit vector $\boldsymbol{n}$. In Fig. 18a it is evaluated from the strain state at the element center and therefore it is more specifically denoted as $\boldsymbol{n}_c$.

In [32], the same function ϕ (based on one single vector $\boldsymbol{n}_c$) was considered for the evaluation of h_b at all Gauss points of the element, but the vector $\boldsymbol{n}$ substituted into (25) was taken at each of the Gauss points separately, as the unit vector $\boldsymbol{n}_c$ in the direction associated with the major principal strain at that point; see Fig. 18b (subscripts $i = 1, 2, 3, 4$ now refer to individual Gauss points). This method will be referred to as Oliver's first method, in short O1. We will also consider a modification, with vector $\boldsymbol{n}$ taken by the same value for all Gauss points, and determined as the unit vector $\boldsymbol{n}_c$ in the direction associated with the major principal strain at the element center; see Fig. 18c. This approach will be referred to as modified Oliver's first method, in short O1m.

Another approach proposed in [32] (in the present paper referred to as Oliver's second method, in short O2) estimates the crack band width as

$$h_b = \frac{A_e}{\boldsymbol{n}_c\cdot\int_{\Gamma_e}\phi\boldsymbol{n}_e\,\mathrm{d}s} \tag{26}$$

where A_e is the element area, Γ_e is the boundary of the element, and $\boldsymbol{n}_e$ is a unit vector normal to the element boundary, oriented outwards, as shown in Fig. 18d. This method uses the same value of h_b for all Gauss points of the element, because the expression in (26) is a global characteristic, evaluated for the whole element. The expression in the denominator of (26) is an equivalent length of the crack segment passing through the element, l_d. Its geometrical meaning is illustrated in Fig. 18d. It is easy to show that the equivalent crack length is equal to the distance between the points

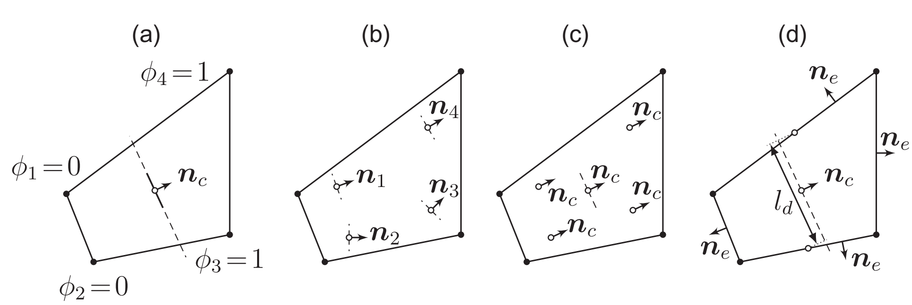

Fig. 18. (a) Determination of nodal values of function ϕ (the hollow circle is the element center), (b) Oliver's first method (the hollow circles are the Gauss points), (c) modified Oliver's first method (the hollow circles are the Gauss points and the element center), and (d) Oliver's second method (the hollow circles are the element center and the midpoints of the sides crossed by an assumed crack segment passing through the center).

obtained by projecting the midpoints of the two element edges crossed by the crack, projected onto the crack direction (perpendicular to $\boldsymbol{n}_c$).

For linear triangular elements, all three Oliver's methods give exactly the same h_b. For bilinear rectangular elements, O1m leads to the same h_b as O2, but O1 gives in general different values of h_b at different Gauss points. For bilinear quadrilaterals, even methods O1m and O2 would in general lead to different results. In the following examples, Oliver's methods will be compared to the projection method (labeled as P) and modified projection method (labeled as Pm).

To better understand the difference between projection and Oliver's approach, Fig. 19 shows the geometrical meaning of the band width estimates constructed for a rectangular element of sides h_x and h_y. Suppose that the major principal direction $\boldsymbol{n}_1$ at the Gauss point under consideration deviates from the horizontal direction by angle α. The projection method proceeds according to Fig. 19a, and the resulting estimate is obtained by projecting the upper element edge under angle α and the left edge under angle $90° - \alpha$; it is thus given by $h_b^{(P)} = h_x \cos\alpha + h_y \sin\alpha$. This formula for rectangular elements was derived in [27]. For Oliver's first method, function ϕ varies linearly from 0 at the left edge to 1 at the right edge; see Fig. 19b. The gradient is thus a constant horizontal vector of magnitude $1/h_x$, its scalar product with the unit vector $\boldsymbol{n}_1$ is $\cos\alpha/h_x$, and the estimated band width according to (25) is $h_b^{(O1)} = h_x/\cos\alpha$. This formula was proposed in [26]. Geometrically, it can be obtained as the length of the inclined segment marked in Fig. 19b. Finally, Oliver's second method is based on the projection of the vertical segment $l_d = h_y \cos\alpha$ (see Fig. 19c), which is then transformed into the band width $h_b^{(O2)} = A/l_c = h_x h_y/(h_y \cos\alpha) = h_x/\cos\alpha$. This is the same result as for Oliver's first method, provided that the principal directions at the Gauss point and at the element center are the same.

Fig. 19d illustrates the difficulties that may arise if the major principal directions at the Gauss point and at the element center are substantially different. The nodal values of function ϕ are always determined by splitting the element from the element center by a segment perpendicular to the major principal direction, $\boldsymbol{n}_c$. According to Oliver's first method, the direction along which the band width is measured is taken as the major principal direction at the Gauss point, e.g. direction $\boldsymbol{n}_2$ at Gauss point 2. Under certain circumstances, this direction can become almost parallel to the isolines of function ϕ (which are vertical in Fig. 19d), and the estimated band width becomes very large, theoretically even infinite if the major principal direction is perfectly aligned with the isolines. This can happen neither for the modified first method, nor for the second method, because they work exclusively with the principal directions at the element center.

6. Numerical results and discussion

6.1. Quadrilateral elements, straight band

The influence of the criterion for estimation of the crack band width is first examined by simulations exploiting the reference mesh in Fig. 8b. The process zone is covered by square elements with bilinear displacement interpolation and 2×2 Gauss integration. The mesh lines are horizontal and vertical, and so the crack band can easily propagate in one vertical layer of elements. The resulting load–displacement curves are plotted in Fig. 20a. The solid curve corresponds to the reference solution with the band width fixed to 5 mm, which is the actual width of the vertical element layer. The modified projection method leads to an almost identical result; see the dashed curve labeled as Pm. Equally good results are obtained with Oliver's second method, which is in this case identical with the modified version of Oliver's first method. The corresponding dotted curve, labeled as O1m = O2, is not detectable in Fig. 20a because it is overlaid by the reference solid curve. On the other hand, the standard projection method results into a lower peak load and reduced energy dissipation; see the dash-dotted curve labeled as P. Oliver's first method (O1) gives almost the same peak load as the reference solution but the post-peak branch is steeper and approaches the solution obtained with the standard projection method.

The deviations from the reference solution are caused by the rotation of principal axes at Gauss points of the cracking elements. The solution is symmetric with respect to the vertical axis of the crack band and, at points located at the axis of symmetry, the principal directions (of strain and stress) are aligned with the mesh lines. The modified projection method and modified Oliver's first method are based on the average strain, for which the principal directions are also aligned with the mesh lines and the resulting estimated crack band width exactly corresponds to the actual width of the vertical element layer. In contrast to that, the standard projection method and standard Oliver's first method deal with the principal directions at each Gauss point separately. For the 2×2 integration scheme, the Gauss points are slightly off the axis of symmetry. As the crack band propagates from the notch upwards, the bottom edge of a newly cracking element is stretched more

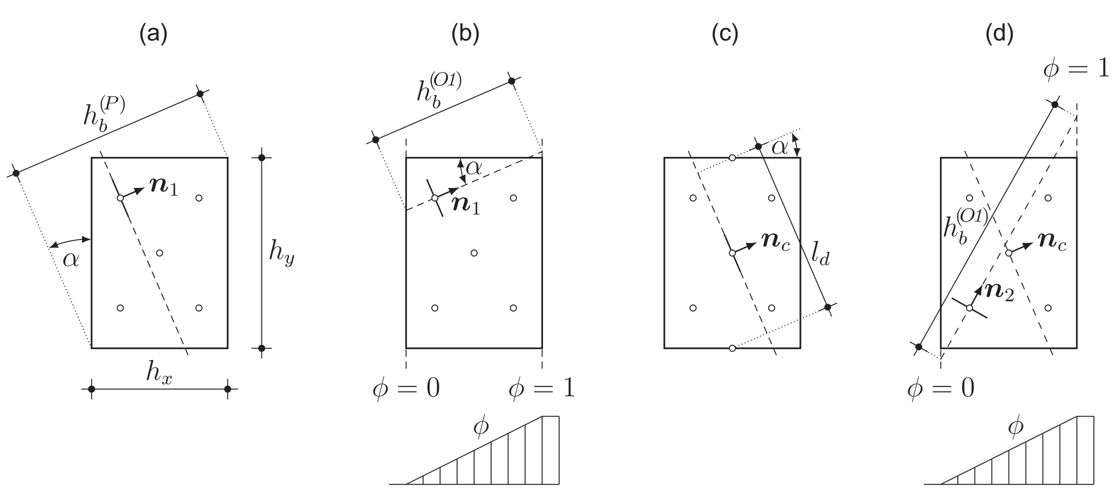

Fig. 19. Band width estimate for a rectangular element: (a) projection, (b) Oliver's first method, (c) Oliver's second method, and (d) situation in which Oliver's first method gives an extremely high band width estimate.

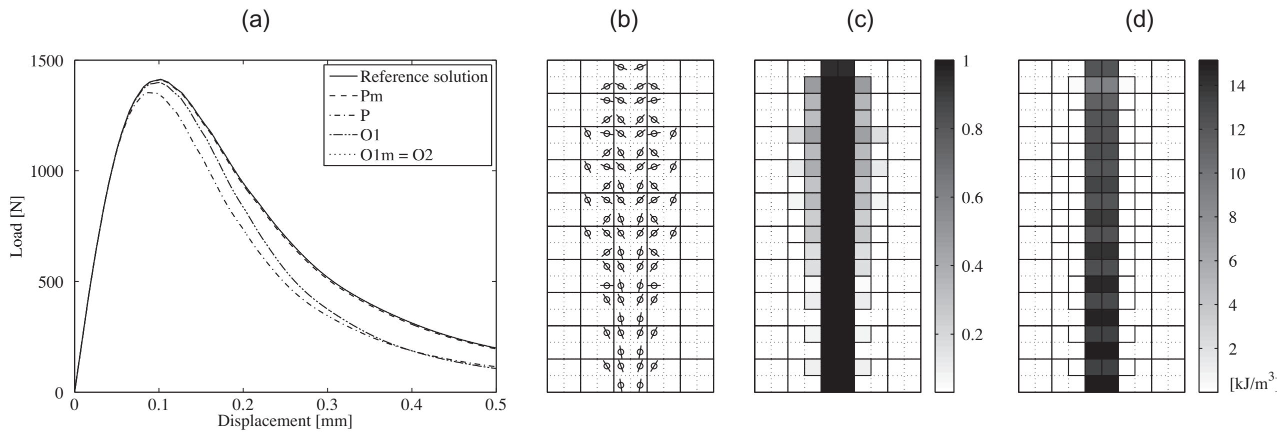

Fig. 20. Bilinear quadrilaterals, reference mesh with full 2×2 integration: (a) load–displacement curves obtained with crack band width estimated by different methods, (b) estimated crack directions at cracking Gauss points, (c) spatial distribution of damage, and (d) spatial distribution of dissipated energy density.

than the top edge, and the bilinear displacement interpolation results into shear strains at Gauss points and, consequently, into rotated principal directions. For the projection method, this is illustrated in Fig. 20b, which shows the assumed crack directions, i.e., segments perpendicular to the major principal strain direction at damage initiation. The displayed rectangular area corresponds to the dotted rectangle above the notch in Fig. 8a. Fig. 20b reveals that the deviation of the assumed crack directions from the vertical direction is quite small in elements just above the notch (i.e., in the bottom part of the figure) but it increases in elements that are more distant from the notch and start cracking later (middle and upper parts). The deviation at certain Gauss points in the middle of the ligament is close to 45°, which results into an important overestimation of the crack band width (by a factor of $\sqrt{2}$). The dissipation per unit volume is reduced as if the crack band were 7 mm wide, but since the actual band width is still just 5 mm, the overall dissipation is underestimated and the response becomes more brittle.

From Fig. 20b it transpires that the cracking affects not just the central layer of elements, but also two additional layers around the central one. However, this does not mean that the localized crack band is three elements wide. Cracks in the elements surrounding the actual crack band (central layer) are initiated when the respective element is approached by the propagating crack band front, but they do not open too much and start closing soon after the front passes by and the cracking process localizes. This is documented in Fig. 20c, which shows that damage levels close to 1 are attained in the central layer only. In this figure, each element is divided into four smaller square boxes, with levels of gray represent the damage at individual Gauss points. Representation in terms of the damage variable could be misleading because the levels of damage outside the central element layer may seem to be important. However, the contribution of these outer layers to energy dissipation is totally negligible, as indicated in Fig. 20d. The energy dissipated in the central layer amounts to 99.48% of the total dissipation. One should bear in mind that for elements much smaller than the characteristic material size (discussed in Section 3), the post-peak branch of the stress–strain diagram is quite flat and intermediate damage levels correspond to very low strength reduction and dissipation. For instance, for the present material parameters and for the band width $h_b = 5$ mm, damage levels of 0.4 or 0.6 correspond to a reduction of the stress by as little as 1.0% or 2.3% (with respect to the peak stress), and to dissipated energy density 189 or 85 times smaller than at complete failure. It is also interesting to note in Fig. 20d that, in typical elements in the central layer, the dissipation is higher in the bottom part of the element and lower in the top part. This is because the deviation of the assumed cracks from the vertical direction is less pronounced at the "bottom" Gauss points of each element than at the "top" Gauss points, as shown in Fig. 20b.

As already mentioned, the modified methods based on the average strain in each element provide, in the present case, a perfect estimate of the actual crack band width. The same effect can be achieved by selectively reduced integration [45], with the normal strains determined at each Gauss point separately but the shear strain evaluated at the element center. For element centers located at the axis of symmetry, the shear strain is zero and the assumed crack direction becomes perfectly vertical; see Fig. 21b. Of course, this holds for the elements in the central layer only, but the neighboring layers give again a negligible contribution to the dissipated energy, as seen in Fig. 21d. For selectively reduced integration, the load–displacement curves obtained with all the band width estimation methods are almost identical; see Fig. 21a. However, when the first Oliver's method is used, the simulation procedure fails shortly after the peak, at the point indicated by the vertical dotted line. This is related to the problem illustrated in Fig. 19d. In the present case, the major principal axis at one Gauss point becomes perpendicular to the major principal axis at the element center, and then the scalar product of vectors $\boldsymbol{n}$ and $\partial\phi/\partial\boldsymbol{x}$ becomes zero and the band width evaluated from (25) tends to infinity. As already mentioned in Section 5.3.3, this kind of problem arises not only if the vectors $\boldsymbol{n}$ and $\partial\phi/\partial\boldsymbol{x}$ are perpendicular but also when they form a large angle, because then the estimated band width h_b becomes too large and may exceed the maximum admissible value $h_{b,\max}$ specified in (14). For the modified version of Oliver's first method, such a problem cannot arise (except if the element is indeed too large).

6.2. Triangular elements, straight band

The reference mesh from Fig. 8b (see also Fig. 10a) is now replaced by a mesh consisting of triangular elements with linear displacement interpolation, constructed by diagonal splitting of each square element of the reference mesh; see Fig. 10c. For constant-strain triangles, there is no difference between the standard and modified projection methods, and all Oliver's methods give the same results. As shown in Fig. 22a, the load–displacement curves obtained with the projection methods (P and Pm) and Oliver's methods (O1, O1m and O2) are very similar and are close to the reference solution. This may seem to be somewhat surprizing, since the assumed crack directions at Gauss points in the crack

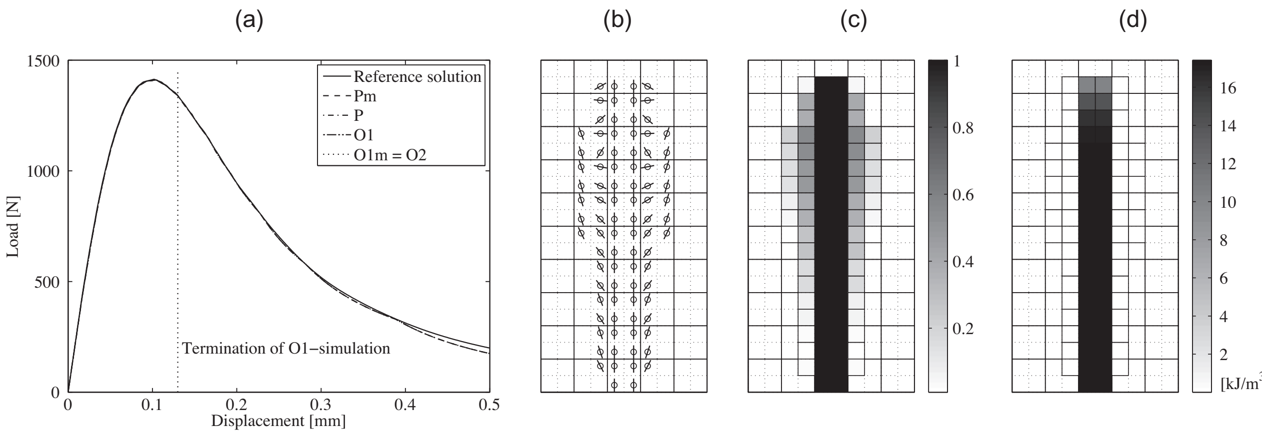

Fig. 21. Bilinear quadrilaterals, reference mesh with selectively reduced integration: (a) load–displacement curves obtained with crack band width estimated by different methods, (b) estimated crack directions at cracking Gauss points, (c) spatial distribution of damage, and (d) spatial distribution of dissipated energy density.

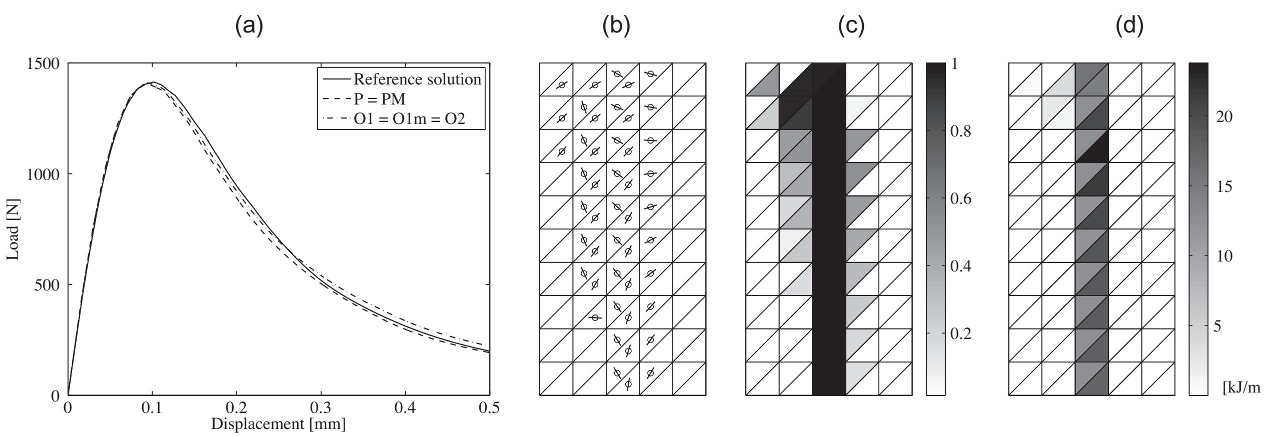

Fig. 22. Linear triangles: (a) load–displacement curves obtained with crack band width estimated by different methods, (b) estimated crack directions at cracking Gauss points, (c) distribution of damage, and (d) distribution of dissipated energy density.

band substantially deviate from the vertical direction (Fig. 22b), and thus the crack band width estimated by the projection methods or by Oliver's methods must differ from the actual value. However, a closer look reveals that, in each pair of triangular elements obtained by diagonal splitting of a square, the assumed crack direction is rotated in the lower right triangle clockwise and in the upper left triangle counterclockwise. Taking into account the element shapes, it follows that the estimated band width of the lower right triangles is smaller than it should be, while for the upper left triangles it is larger. The density of energy dissipation in the lower right triangles is then higher than the optimal value, while in the left triangles it is lower; see Fig. 22d. On the average the errors nearly cancel and the overall dissipation is captured quite accurately.

6.3. Quadrilateral elements, zig-zag band

So far we have considered cases in which the crack band can propagate along the mesh lines, in one straight layer of elements. It is interesting to check how different criteria perform if the band has to cross the mesh in a zig-zag pattern. The central region of the mesh in Fig. 10b consists of square elements with bilinear displacement interpolation, with the mesh lines running at 45° with respect to the expected overall direction of the crack band. Note that, in order to resolve the shape of the notch, three elements just above the notch have somewhat irregular shapes (see the details of the mesh in Fig. 24a).

The mesh is symmetric, but since the crack band propagates in a zig-zag pattern similar to Fig. 17b, the localized solution cannot remain symmetric. The simulation provides a solution with the crack band slightly shifted to the right with respect to the axis of symmetry, as seen in Fig. 23c. In the lower part of the process zone, the assumed crack directions at individual Gauss points (for full 2×2 integration) deviate from the vertical direction only slightly; see Fig. 23b. In the upper part the deviations are much more pronounced and at certain points get close to 90°. For full integration, the load–displacement curves obtained with all methods overestimate the peak load; see Fig. 23a. This can be explained by the special mesh structure in the region just above the notch, where cracking initially starts in two symmetrically located quadrilaterals. Localization into one of these elements (in the present simulation, into the right one) occurs only later, but initially the cracking band actually extends over the width of two elements, which makes the early response more ductile and increases the peak load. For most methods, the post-peak response is somewhat more brittle than in the reference case. The most brittle response is obtained with Oliver's first method, while the other methods remain quite close to the reference curve. An exception is of course the method based on the square root of element area which, as expected, grossly underestimates the actual band width and thus overestimates the energy dissipation; see the dotted curve in Fig. 23a.

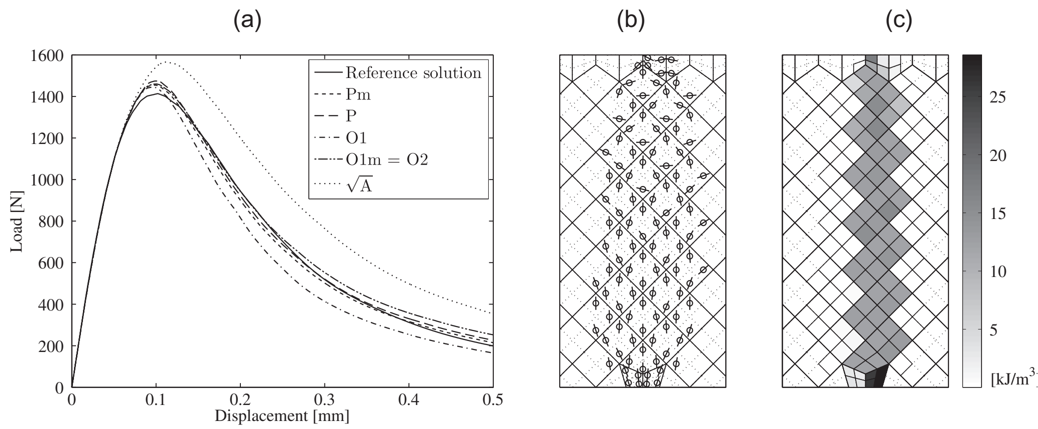

Fig. 23. Bilinear quadrilateral elements with full integration and inclined mesh lines: (a) load–displacement curves obtained with crack band width estimated by different methods, (b) estimated crack directions at cracking Gauss points (for projection method), and (c) spatial distribution of volumic dissipation density (for projection method).

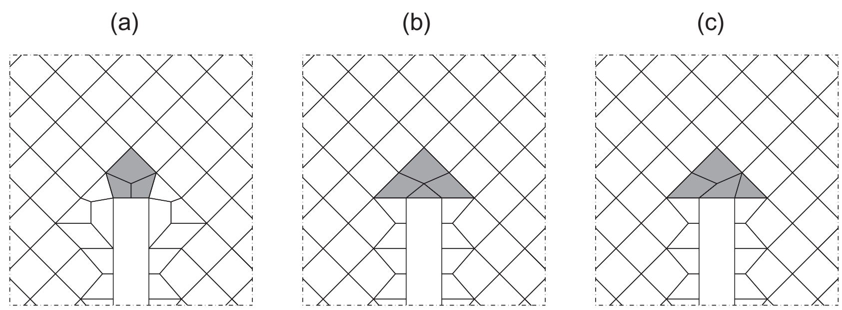

Fig. 24. Quadrilateral meshes with inclined mesh lines and different arrangements of elements above the notch: (a) mesh A from Fig. 10b, (b) mesh B (with one triangular element), and (c) mesh C (with one triangular element).

To get more insight into the role of the specific mesh structure, the simulations have been repeated on meshes with the same inclined mesh lines as in Fig. 10b, but with modified details of the irregular arrangement of elements just above the notch. A close-up of the original mesh from Fig. 10b (further referred to as mesh A) in the area near the notch tip is shown in Fig. 24a. Parts (b) and (c) of Fig. 24 show two locally modified meshes, further referred to as meshes B and C. The load–displacement curves obtained with the projection method and full integration for meshes A, B and C are plotted in Fig. 25a. It is clear that the local details of the mesh above the notch tip have quite a strong influence on the peak load but in the post-peak branch the influence gradually fades away and all curves become almost identical. The highest peak load is obtained for mesh B and the lowest for mesh C, with a relative difference of 7.6%. The source of the discrepancy is revealed in Fig. 26, which shows the spatial distribution of dissipated energy density for the three meshes. Mesh B is symmetric and the actual crack band near the notch is wider than the individual elements; see Fig. 26b. This leads to an excessive dissipation in this area, as confirmed by the distribution of surfacic dissipation density along the ligament; see the solid and dashed lines in Fig. 25b. On the other hand, mesh C is nonsymmetric and the specific arrangement of elements allows for propagation of the crack band in one zigzag layer; see Fig. 26c. The resulting dissipation per unit area is the same as in regular parts of the mesh and is very close

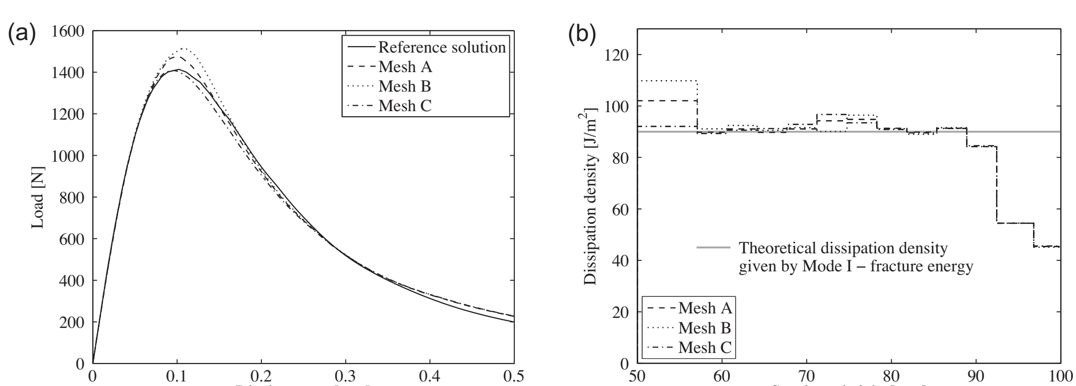

Fig. 25. Influence of the element arrangement near the notch tip on the (a) load–displacement curve and (b) distribution of surfacic dissipation density along the ligament.

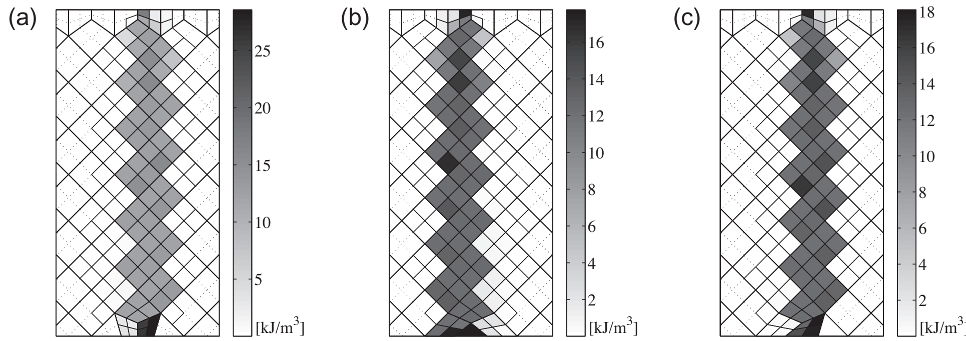

Fig. 26. Influence of the element arrangement near the notch tip on the distribution of volumic dissipation density.

to the fracture energy; see the dash-dotted line in Fig. 25b. The peak load for mesh C is also very close to the reference value.

The graphs in Fig. 25b also indicate that the correct amount of energy is dissipated in the regular part of the zig-zag crack band. For mesh C, one can see an almost perfect match up to the distance of 38 mm from the notch, with the exception of a short segment in which the surfacic dissipation density exceeds the fracture energy by approximately 7%. This is caused by the deviation of the principal directions in this area; see Fig. 23b. The additional orientation factor γ proposed in [43] would reduce the dissipation by up to 33% and the response would become excessively brittle. Thus the need for adjustment of the band width by this factor is not confirmed by the present study.

For selectively reduced integration and mesh A, the load–displacement curves obtained with the standard and modified projection methods get quite close to the reference curve; see Fig. 27a.

6.4. Band along the axis of symmetry

In problems exhibiting symmetry, computational effort can be reduced by simulating only a part of the investigated body, for instance only the left half of the present beam, which is symmetric with respect to the vertical axis. However, one needs to be careful when addressing problems with localized solutions. First of all, strain localization can in some cases break the symmetry. This could happen for instance for a specimen with two symmetrically located notches or initial cracks. The symmetric solution with two propagating cracks may become unstable, and the stable solution with one crack propagating and the other one arrested is then nonsymmetric and cannot be captured on a reduced mesh. For the present case with one notch located on the axis of symmetry, a symmetric solution can be expected and the simulation on a reduced mesh is justified.

In addition to imposing the appropriate boundary conditions on the axis of symmetry, one needs to take into account that a crack band in the layer of elements adjacent to the axis of symmetry corresponds to a crack band of a double width when the solution is mirrored to the other half of the specimen. Therefore, only one half of the actually dissipated energy should be attributed to this "half band". This can be properly handled by assigning special material properties to the elements adjacent to the axis of symmetry, with the same strength but with fracture energy reduced to one half. If this is not done, the response becomes excessively ductile, as demonstrated by the dashed load–displacement curve (labeled as "Original G_F") in Fig. 28. The simulation has been performed on a mesh consisting of square elements that correspond to the reference mesh, but with the central layer split into half by a vertical cut; see Fig. 10d. The elements forming the crack band are thus rectangles of width 2.5 mm and height 5 mm, and the band width has been set to h_b = 2.5 mm. If the simulation is repeated with the fracture energy of this layer reduced to one half (i.e., to 45 J/m^2), the resulting curve perfectly matches the reference solution; see the dash-dotted curve (labeled as "Reduced G_F") in Fig. 28.

After proper adjustment of the fracture energy in elements adjacent to the axis of symmetry, the influence of the band width estimation method can be assessed. The load–displacement diagram in Fig. 29a reveals that the response is too brittle for all

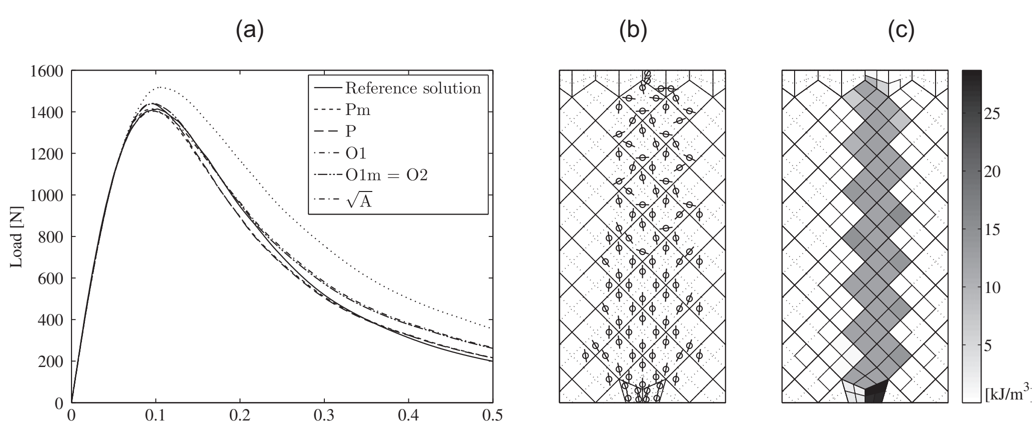

Fig. 27. Bilinear quadrilateral elements with selectively reduced integration and inclined mesh lines: (a) load–displacement curves obtained with crack band width estimated by different methods, (b) estimated crack directions at cracking Gauss points (for projection method), (c) spatial distribution of volumic dissipation density (for projection method).

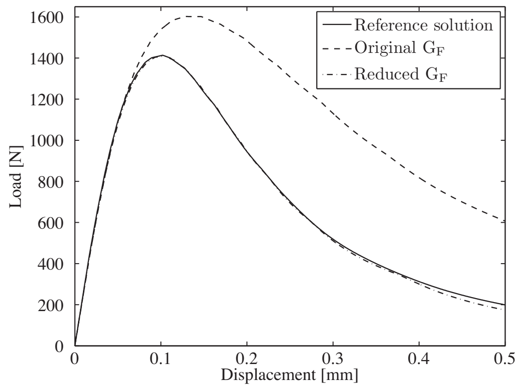

Fig. 28. Bilinear quadrilateral elements with straight crack band and symmetry conditions: load–displacement curves obtained with the same material properties in all elements (original G_F) and with reduced fracture energy in the elements adjacent to the axis of symmetry (reduced G_F).

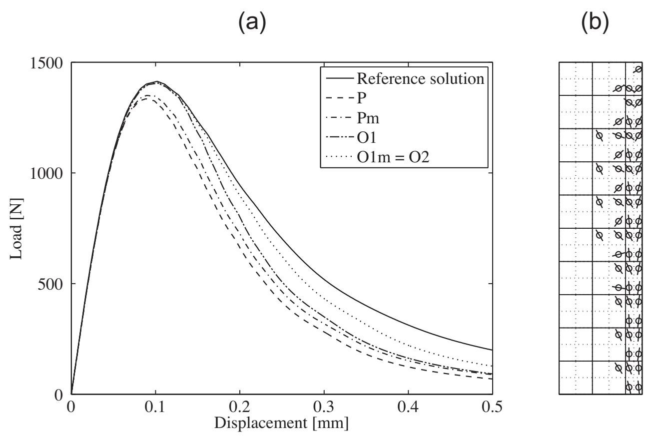

Fig. 29. Bilinear quadrilateral elements with full 2×2 integration, straight crack band, symmetry conditions and reduced fracture energy in the elements adjacent to the axis of symmetry: (a) load–displacement curves obtained with crack band width estimated by different methods and (b) crack directions in the left half of the process zone.

methods considered here. The largest deviation from the reference curve is detected for the projection method (dashed curve) and for its modified version (dash-dotted curve). Oliver's methods give the correct peak load but the post-peak behavior is again too brittle, especially for the first method. The modified first method (in this case equivalent to the second method) leads to the best results; see the dotted curve in Fig. 29a. For selectively reduced integration, the results improve only slightly, and Oliver's first method fails shortly after the peak; see Fig. 30. Deterioration of the results compared to the simulation on the full mesh is of course due to the fact that the centers of cracking elements are not on the axis of symmetry and the principal axes are rotated (Fig. 29b), which leads to an overestimated band width and underestimated dissipation density.

6.5. Higher-order elements

Certain fundamental difficulties related to the resolution of localized solutions by higher-order elements were illustrated by one-dimensional examples in Section 4. It was shown that, in the one-dimensional setting, the correct overall energy dissipation can be enforced by a proper adjustment of the crack band width, affected not only by the element size but also by the integration scheme. In particular, the appropriate band width turned out to

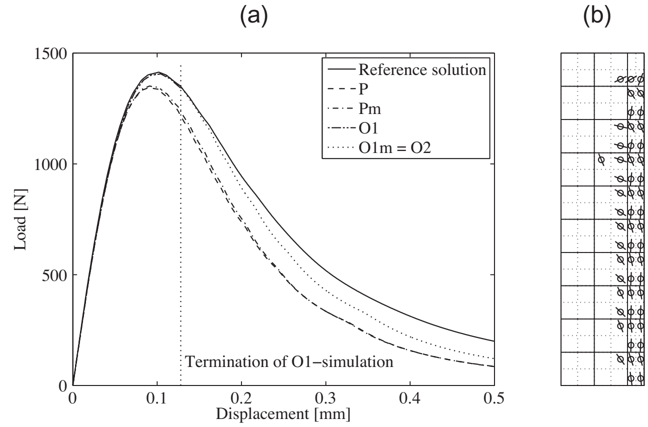

Fig. 30. Bilinear quadrilateral elements with selectively reduced integration, straight crack band, symmetry conditions and reduced fracture energy in the elements adjacent to the axis of symmetry: (a) load–displacement curves obtained with crack band width estimated by different methods and (b) crack directions in the left half of the process zone.

be one half of the element size for two-point integration and 13/18 times the element size for three-point integration. Simulations of the bending beam using eight-node square elements (with a serendipity-type quadratic interpolation of displacements) indicate that such rules derived from one-dimensional analysis may lead to satisfactory results if the crack band is aligned with the mesh but not in a general case.

Fig. 31a shows the load–displacement curves obtained on the reference mesh (with quadratic elements and with additional nodes placed at midpoints of each element side). The dotted and dash-dotted curves that are far below the reference curve correspond to the standard evaluation of the crack band width by projecting the element, while the other two curves that are just below the reference curve have been constructed by simulations with the element size multiplied by 1/2 for 2×2 integration and by 13/18 for 3×3 integration. As shown in Fig. 31b, for 2×2 integration the cracking process in quadratic elements indeed localizes into one layer of Gauss points and not into one layer of finite elements, and so the effective width of the cracking band is 1/2 of the element size. For the mesh from Fig. 10b, with mesh lines inclined by 45° and with quadratic elements, the results are much less satisfactory. The cracking band in Fig. 32b is not a clean zigzag pattern and too much energy is dissipated, which is confirmed by the dashed curve in the load–displacement diagram in Fig. 32a. For 3×3 integration, the response is too brittle if the entire element size is used and too ductile if the correction factor of 13/18 is applied.

7. Summary and conclusions

Examples presented in this paper and their detailed analysis have revealed that practical application of the crack band approach as a technique eliminating the pathological mesh sensitivity is not straightforward. The success depends on many factors that the users of finite element simulation tools are not always aware of. A striking example is the criterion for estimation of the width of the crack band. The band width is often considered simply as the "element size" and its evaluation in many commercial codes is based on the square root of element area in two dimensions or cubic root of element volume in three dimensions. However, the actual effective width of the localized crack band that forms in a finite element analysis based on a material law with softening is affected by the element type, element shape, direction of the crack

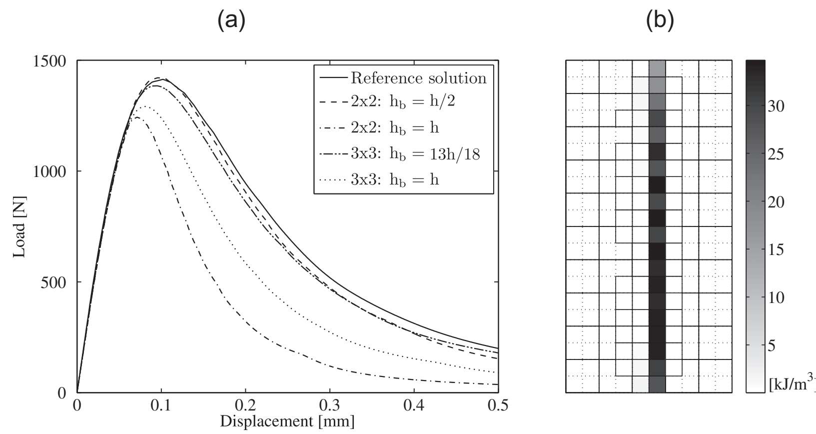

Fig. 31. Quadratic quadrilateral elements with mesh lines aligned with the crack band: (a) load–displacement curves for different integration schemes and different criteria regarding the effective element size and (b) spatial distribution of dissipated energy for 2×2 integration and $h_b = h/2$.

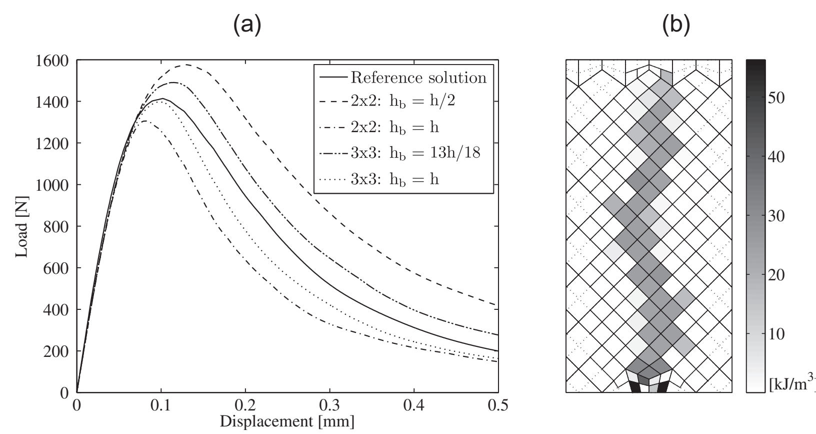

Fig. 32. Quadratic quadrilateral elements with mesh lines inclined with respect to the crack band: (a) load–displacement curves for different integration schemes and different criteria regarding the effective element size and (b) spatial distribution of dissipated energy for 2×2 integration and $h_b = h/2$.

band with respect to the mesh lines and, for higher-order elements, even by the integration scheme. The simple "square-root-of-area" criterion can lead to gross errors, comparable to a misprediction of the fracture energy by 50% or even more. Additional potential sources of error are hidden in the choice of the specific ingredients of the constitutive model (e.g., of the equivalent strain expression for strain-driven isotropic damage models), or in the reduction of the mesh justified by symmetry considerations. The following comments, hints and recommendations can contribute to the reduction of such errors:

- Constitutive models that give the same response under uniaxial tension can give different responses in the presence of lateral stress. Since the actual stress state in front of a propagating crack is often biaxial, the amount of dissipated energy can be affected by the specific constitutive formulation. Therefore, matching the fracture energy under uniaxial tension does not guarantee that the same amount of energy is dissipated in a propagating crack band, even under pure mode-I conditions.
- In contradiction to the common belief that the minimum size of the localized zone in one-dimensional failure simulations is the size of one finite element, numerical solutions localized into sub-element domains can be detected in simulations with higher-order elements. In the one-dimensional setting, this effect can be taken into account by a proper adjustment of the formula for the effective band width, such that the global energy dissipation is captured correctly. However, the corresponding stress field is highly disturbed and a large error even in terms of the global dissipation can be induced in multiple dimensions, especially if the crack band is not aligned with the mesh lines. Therefore, higher-order elements are not suitable for crack band simulations, and the simplest (multi)linear elements should be preferred.
- The effective width of a crack band depends on the element shape and orientation (with respect to the overall direction of the band). Simple criteria based exclusively on the element area or volume can lead to large errors and thus cannot be recommended.
- The method estimating the effective band width by projecting the element onto the major principal strain axis gives superior accuracy if the principal strain is taken at the element center (or as the average) rather than at each Gauss point separately. The need for an additional orientation factor proposed in [43] has not been confirmed.
- Among the methods proposed by Oliver [32], the second method, based on formula (26), gives in general better results than the first method, based on formula (25). If the direction $\boldsymbol{n}$ in formula (25) is determined from the strain at the element center (or from the average strain), the first method gives results very similar to the second method, and for some regular element shapes both methods become equivalent. In the original form, with direction $\boldsymbol{n}$ evaluated from the strain at each

Gauss point separately, the first method may result into excessively large or even infinite estimates of the crack band width. The second method is thus superior.

- There is no need for the additional "orientation factor" γ proposed in [43]. Introduction of this factor would lead to an unrealistic reduction of dissipated energy, especially for crack bands inclined with respect to the mesh lines.
- If the simulation takes advantage of symmetry and only one half of a symmetric domain is discretized, the elements adjacent to the axis of symmetry need to be assigned to a special material with fracture energy reduced to one half (or the band width computed for these elements needs to be doubled). If a crack band develops in the layer of elements adjacent to the axis of symmetry, accuracy of the band width estimate is reduced by the fact that the element centers are not on the axis of symmetry.

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